Benha University

Faculty of Engineering

Mechatronics, Systems and AI Automation Program

AI Lab — Project 1E: Course Study Assistant

StudyMate

A Retrieval-Based Course Study Assistant that Answers Only from the Lecture Slides

FINAL PROJECT REPORT

Prepared by	Ahmed Hallawa (Ahmed Esam Yahia)
Course	Artificial Intelligence Laboratory (AI Lab)
Submitted to	Course instructor and teaching assistant
Live system	https://ahmedhallawa.pythonanywhere.com
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Abstract

This report presents **StudyMate**, a web-based study assistant developed for the AI Lab course (Project 1E). Students usually search dozens of lecture slides by hand to answer a single question, while general-purpose AI chatbots produce fluent answers that may contradict the lecture and give no reference to verify. StudyMate addresses this with an information-retrieval approach: the lecture PDFs are split into one passage per slide, indexed with TF-IDF, BM25 and Latent Semantic Analysis, and every answer is an actual slide shown with its lecture and slide number. A tuned rejection gate returns *Not found in the course* for questions the slides do not cover, and a Logistic Regression classifier detects the topic of each question to drive a per-topic mastery model and study recommendations.

The system was developed iteratively (64 commits) from a single-course Flask prototype into a deployed, bilingual (English/Arabic) platform with three roles — student, doctor (lecturer) and administrator — each with its own workflow and permissions. It includes a Study Studio (quiz, mind map, videos, exams, progress), automatic course creation from uploaded PDFs, an automatic per-course evaluation, and a questions follow-up for lecturers. On the hand-labelled test sets the topic classifier reaches 88.9% accuracy, slide search reaches 80.0% Top-3, and all off-topic questions are rejected.

Keywords: information retrieval, TF-IDF, BM25, latent semantic analysis, text classification, intelligent tutoring, Flask, bilingual interface

Key results

Table 1. Key results

Item	Result
Topic classifier (Logistic Regression)	88.9% accuracy · macro-F1 88.9%
Slide search, hand-made test (30 questions)	Top-1 70.0% · Top-3 80.0% (Hybrid)
Off-topic questions refused	10 / 10
Search methods available	TF-IDF · BM25 · LSA · Hybrid · Neural (optional)
Courses / doctors	unlimited courses, each owned by a doctor (≤10 per doctor)
Size of the system	64 web routes · ~12,236 lines of code · 70 tests (all passing)
Interface	English / Arabic (RTL), light / dark, desktop and phone

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CHAPTER 1

The Problem

1.1 Background

In engineering courses the lecture slides are the main reference: one course easily has 150–300 slides spread over 7–12 PDF files. Before a quiz or exam a student has a specific question ("what is an admissible heuristic?") and has to find the slide that explains it.

1.2 Problem statement

Table 2. The problems StudyMate addresses

Problem	Effect on the student / doctor
Searching slides by hand is slow	PDF search (Ctrl+F) needs the exact word; a question in the student's own words finds nothing.
General chatbots hallucinate	Answers are fluent but may contradict the lecture and give no slide reference to check.
Questions in Arabic, slides in English	Many students think and ask in Arabic.
No picture of the student's level	The student does not know which topic to study next; the doctor does not know which topics the class finds hard.
Questions get lost	Questions asked outside the lecture are never collected or answered.
Material is scattered	PDFs, recorded lectures, YouTube links and exams live in different places.

1.3 Project requirements (AI Lab, Project 1E)

Table 3. Requirements vs. delivery

Requirement	Delivered
Flask app with Ask, Practice, Progress, Admin, Evaluation, Dashboard	All pages, plus Studio, exams, badges, chat, calendar, help, videos
TF-IDF + cosine search (BM25 bonus)	TF-IDF, BM25, LSA, Hybrid, Neural
Topic classifier (NB or LogReg)	Both trained and compared; LogReg selected
Mastery update + "study next"	Per-topic level after each answer
≥8 topics, ≥5 questions/topic, 300 labelled questions, 30 search tests, 10 off-topic, data card	12 topics, 72 bank questions, 360 labelled, 30 search, 10 off-topic, data_card.md
logs.xlsx written by one function	`log_to_excel` → Questions / Answers / Visits / Feedback
Evaluation: accuracy, confusion matrix, per-class P/R/F1, top-1/top-3, off-topic, robustness	All, plus automatic per-course evaluation
Bonus: deployment, Arabic	Deployed on PythonAnywhere; full Arabic UI and Arabic questions

CHAPTER 2

The Solution

2.1 Idea

StudyMate is a **retrieval** system, not a generator: every answer is a real slide. The slides are split into passages (one per slide), indexed, and a question is matched against them. A tuned "Not found" gate refuses questions the slides do not cover. A topic classifier labels each question, which drives the student's per-topic level, the doctor's dashboard and the practice questions.

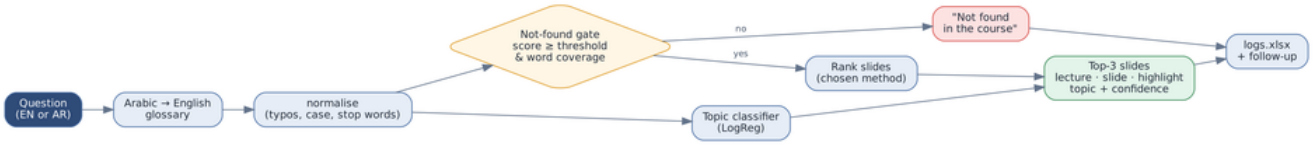


Figure 1. How a question is answered: translation, normalisation, Not-found gate, ranking and topic detection.

2.2 System architecture

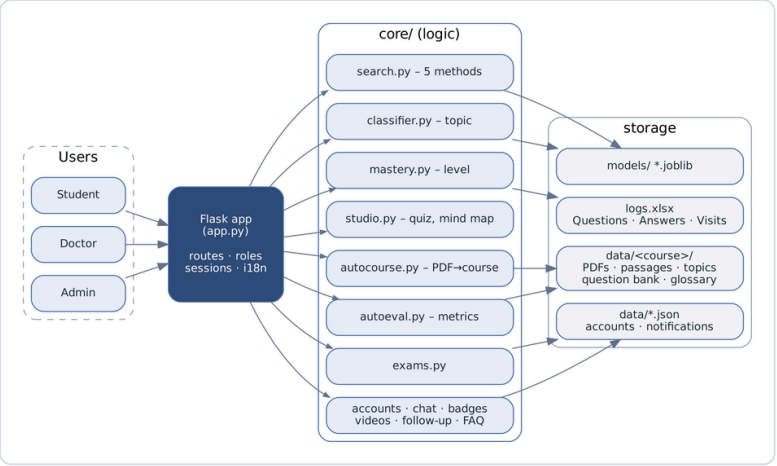


Figure 2. Architecture: one Flask app, logic modules in core/, file-based storage.

Flask + Jinja serve every page; the session holds the role (student, doctor, admin).

core/ holds the logic: search, classifier, mastery, Study Studio, exams, automatic course building and evaluation, accounts, chat, badges, videos and follow-up.

Storage is file based, which suits free hosting: each course in `data/<code>/`, trained models in `models/`, the activity log in `logs.xlsx`, and accounts / notifications in JSON files that are never committed to git.

2.3 Data

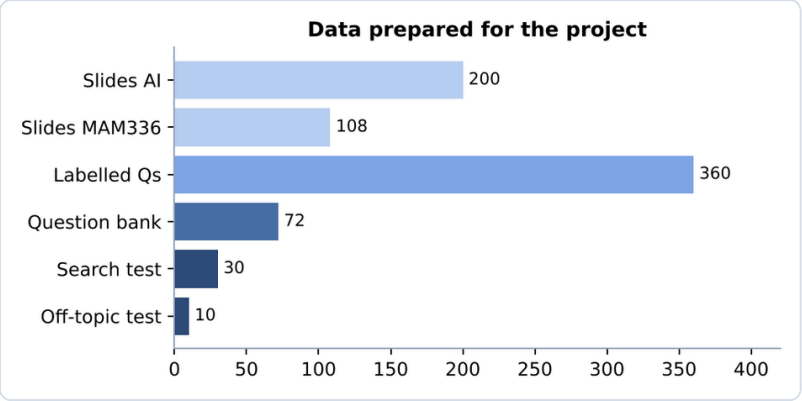


Figure 3. Data prepared for the AI course (and the slides of MAM336).

- **Slides:** 7 AI lectures → 200 passages (pypdf); 5 Automotive (MAM336) lectures → 108 passages, partly OCR because the PDFs are scanned.
- **12 topics** with a topic map (lecture/slide → topic).
- **360 labelled student-style questions** for the classifier, **72** practice questions with two hints each.
- **30** search questions with the correct slide, **10** off-topic questions.
- An **Arabic→English glossary** and a one-page **data card** (source, size, labels, privacy).

2.4 Models used and how they were trained

Table 4. Models in StudyMate

Model	Purpose	Training / settings
TF-IDF + cosine	Default slide search	uni+bi-grams, sublinear tf, English + custom stop words, max_df 0.6
BM25 (Okapi)	Probabilistic ranking	rank_bm25 on the same tokens, rescaled to the cosine range
LSA	Semantic search (related words)	TruncatedSVD, 100 dimensions on the TF-IDF matrix
Hybrid	Exact + related words	0.5 × TF-IDF + 0.5 × LSA
Neural (optional)	Free sentence-embedding model	fastembed BAAI/bge-small-en-v1.5, 0.7 neural + 0.3 TF-IDF; falls back to Hybrid when not installed
Not-found gate	Refuse off-topic questions	best TF-IDF score ≥ 0.11 and question-word coverage ≥ 0.3, tuned on the off-topic set
Naive Bayes / Logistic Regression	Topic of the question	TF-IDF features; stratified 80/20 split of 360 questions (72 test); LogReg selected
Mastery model	Level per topic	level ← level + 0.35·(result − level); result 1 / 0.7 / 0.4 / 0 for 0 / 1 / 2 hints / wrong
CNN (Colab notebook)	Recognise lecture images	MobileNetV2 transfer learning on Colab (small CNN offline), 128-d embedding for image→slide search

Training is reproducible with three scripts: `scripts/build_passages.py` (PDF → passages), `scripts/train.py` (classifiers + search index) and `scripts/evaluate.py` (all metrics, figures and the threshold). When a doctor uploads a new course the same steps run automatically on the server (`core/autocourse.py`).

2.5 Evaluation results

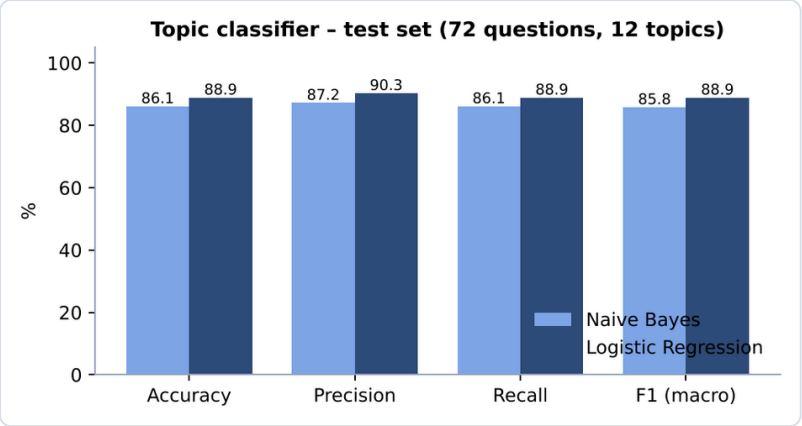


Figure 4. Topic classifier on the held-out test set.

Both classifiers were trained on the same TF-IDF features. **Logistic Regression** wins on every measure (accuracy 88.9%, macro-F1 88.9%) and is the model used by the site.

Most errors are between neighbouring topics (e.g. *Heuristics* vs *A* search*), which share vocabulary.

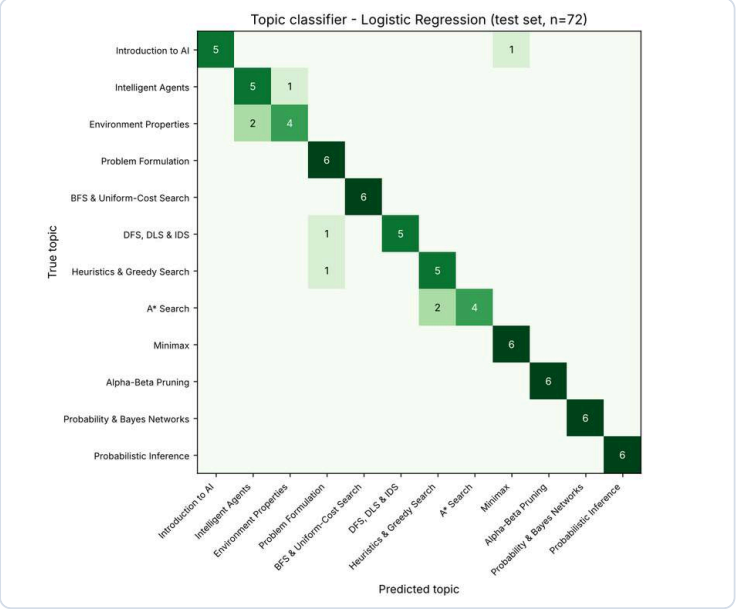


Figure 5. Confusion matrix of the Logistic Regression classifier.

Rows are the true topic and columns the predicted topic. The strong diagonal shows that almost every topic is recognised correctly; the few off-diagonal cells are the confusions mentioned above.

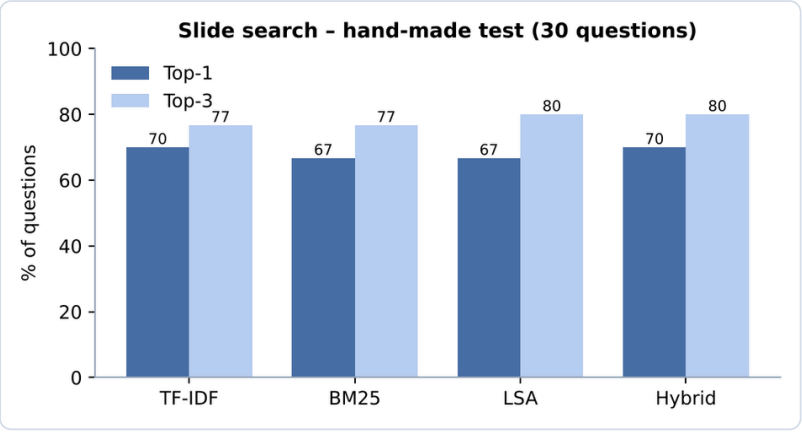


Figure 6. Slide search on the hand-made test set (30 questions with the correct slide).

Top-1 = the correct slide is the first result; **Top-3** = among the three shown.

TF-IDF and Hybrid reach 70.0% Top-1; the semantic methods (LSA, Hybrid) give the best Top-3 (80.0%). All 10 off-topic questions were refused.

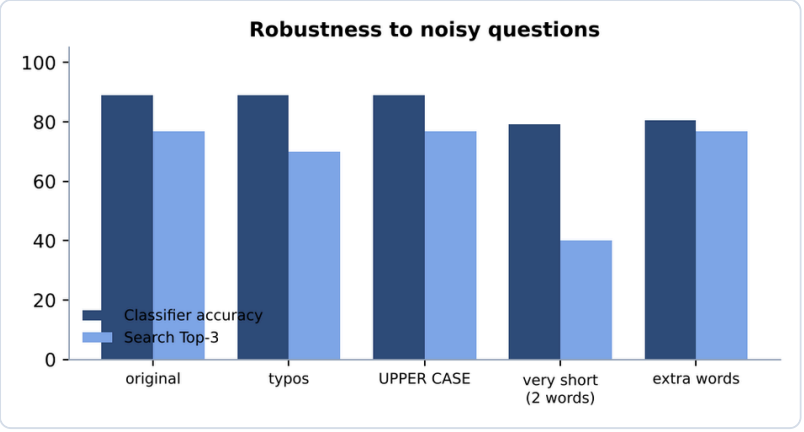


Figure 7. Robustness: the same questions with typos, capitals, only two words, extra words.

Typos, upper case and extra words barely change the results. Very short questions (two key words) are the weak point: Top-3 drops to 40 % because two words rarely identify one slide. Arabic test questions were all classified correctly through the glossary.

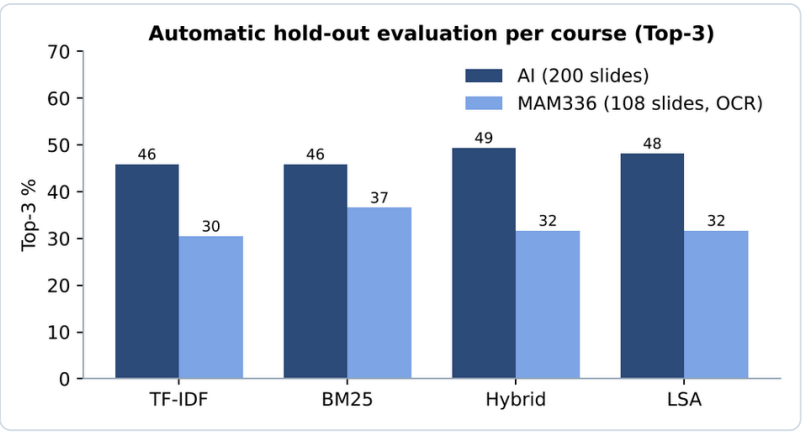


Figure 8. Automatic hold-out evaluation, run again after every upload (Top-3).

For a new course there is no hand-made test set, so the site removes one sentence from every slide, rebuilds the index without it and asks it as a question. This is harder than the hand test, so numbers are lower (AI Top-3 45.9%–49.4%).

MAM336 is lower because its scanned PDFs give noisy OCR text — the evaluation makes such data problems visible to the admin.

CHAPTER 3

Development Journey

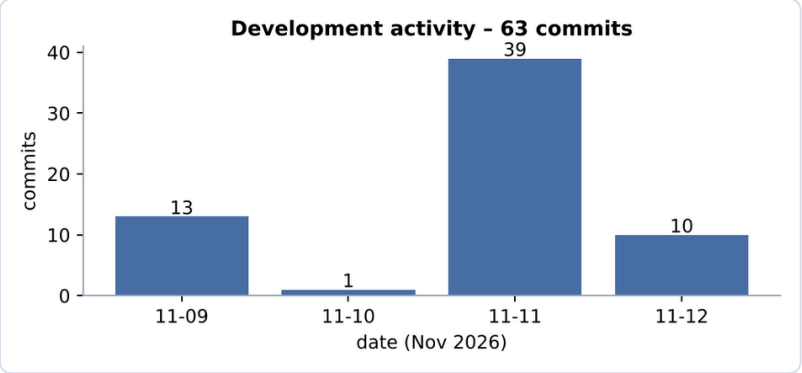


Figure 9. Commits per day on GitHub.

The project was built in short, tested iterations: 64 commits, each one pushed to GitHub and then pulled on PythonAnywhere. Every iteration was driven by testing the live site and by feedback screenshots.

3.1 How the interface evolved

The screenshots below were taken by running the exact historical versions from the git history.

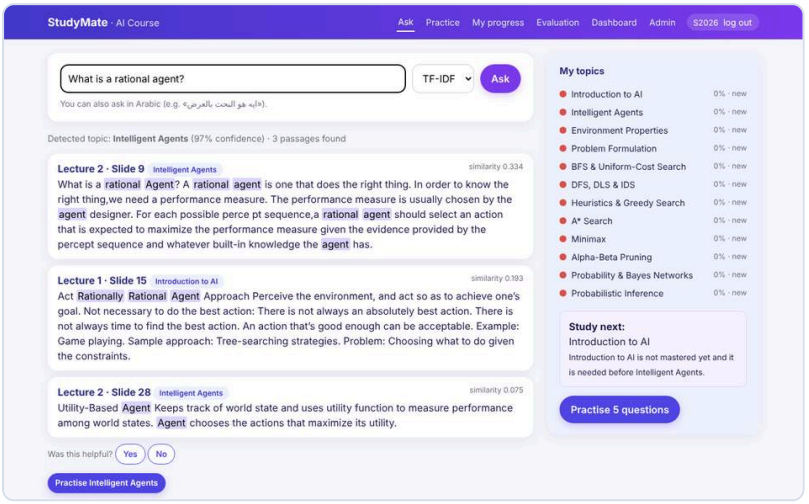


Figure 10. v1 · 9 Nov — first Flask prototype

v1 · 9 Nov — first Flask prototype

Purple header with text links (Ask, Practice, My progress, Evaluation, Dashboard, Admin). Search returns the top-3 slides with the topic; one course; the lecturer area is behind one shared password.

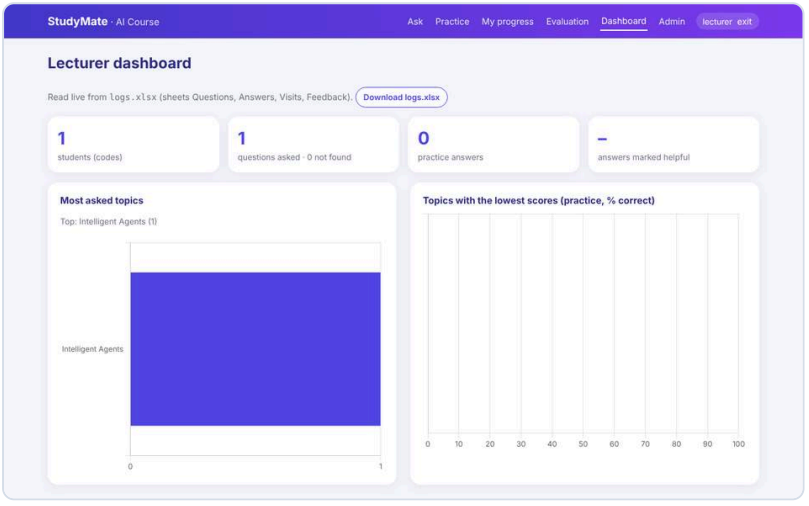


Figure 11. v1 · lecturer dashboard

v1 · lecturer dashboard

Read live from logs.xlsx: number of students, questions, practice answers and charts of the most asked / weakest topics.

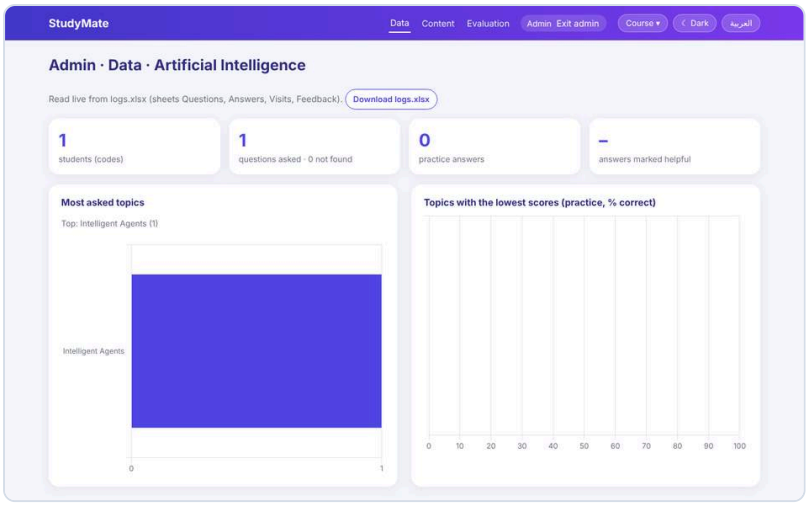


Figure 12. v3 · 9 Nov — Arabic/English, dark mode, several courses, doctors

v3 · 9 Nov — Arabic/English, dark mode, several courses, doctors

The admin area is separated from the student area; doctor accounts with private codes; a course drop-down; the Automotive course and guest mode were added.

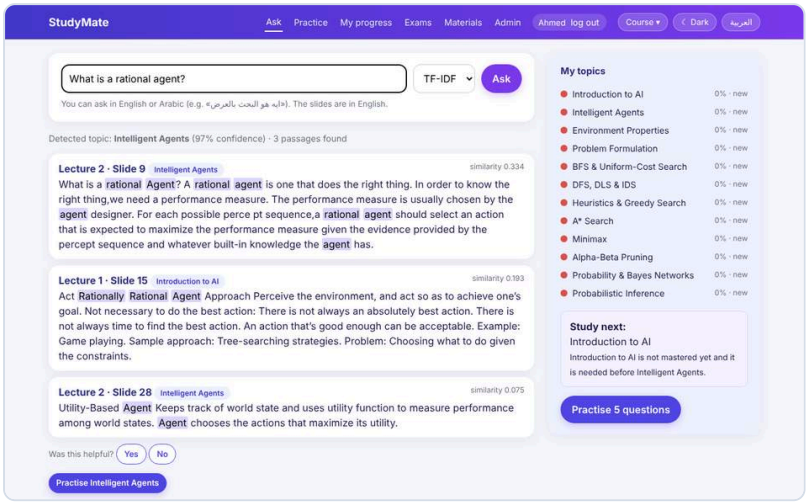


Figure 13. v4 · 11 Nov — accounts and exams

v4 · 11 Nov — accounts and exams

Real student accounts (register, forgot password, remember me), exams and materials appear in the navigation; Arabic translation of answers.

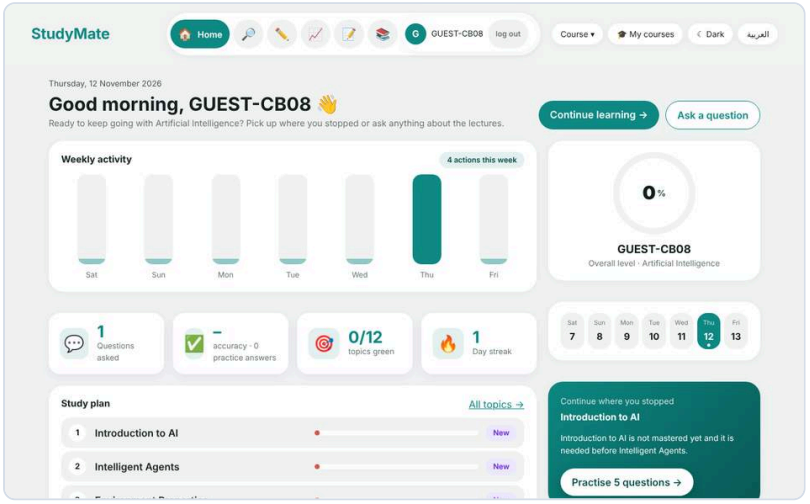


Figure 14. v5 · 11 Nov — new "Edu" design

v5 · 11 Nov — new "Edu" design

Complete redesign: light dashboard, round icon navigation capsule, student home with weekly activity, overall level, calendar strip and study plan (teal palette).

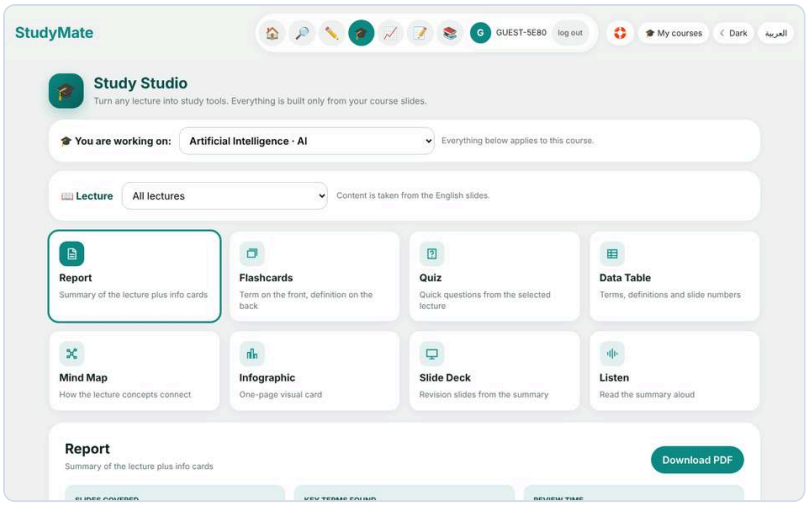


Figure 15. v6 · 12 Nov — first Study Studio

v6 · 12 Nov — first Study Studio

Eight study tools built from the slides (report, flashcards, quiz, table, mind map, infographic, slide deck, listen). Later reduced to the ones students really used.

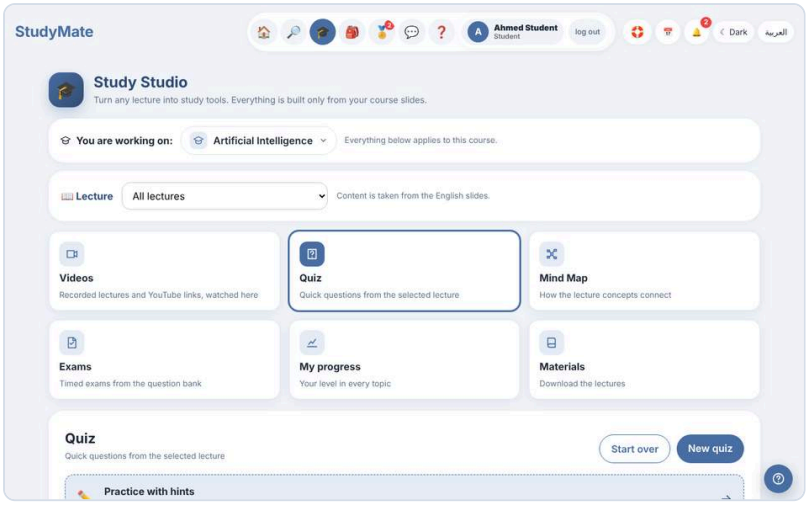


Figure 16. Final · 12 Nov — slate-blue design

Final · 12 Nov — slate-blue design

Calmer slate-blue palette, one set of line icons, Studio with Videos / Quiz / Mind map / Exams / Progress / Materials, course drop-down with icons, floating help button.

3.2 Problems we met and how they were solved

Table 5. Main problems during development

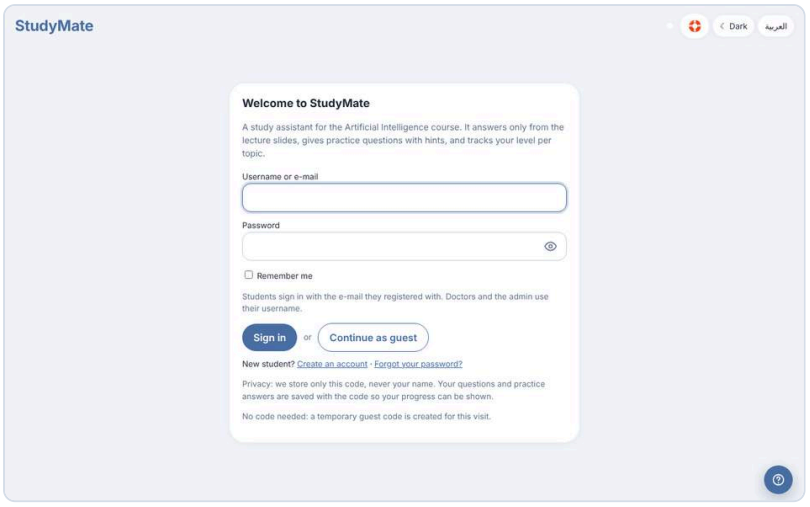
Problem	Cause	Solution
Broken words in slide text	pypdf glues words and symbols from slide layouts	Text cleaning + word splitting (wordninja)
Scanned PDFs (MAM336)	Slides are images, no text	OCR with a cache per page; evaluation shows the lower quality
Wrong answers to off-topic questions	Cosine always returns something	Threshold + word-coverage gate tuned on the off-topic set (10/10 refused)
Arabic questions	Slides are English	Arabic→English glossary before searching
Admin and student pages mixed	One session for both	Separate areas; a signed-in doctor/admin never sees student pages
Doctors seeing each other's data	Courses had no owner	Owner per course; a doctor manages and sees only his courses and students
Slow pages	Excel read/write on every request	Visits written by a background thread; Excel reads cached by modification time
Password guessing	Unlimited login attempts	5 tries per account, then a support pop-up
No e-mail on free hosting	Outbound internet is an allow-list	Admin resets a locked account from the Support inbox; strong generated passwords
Deploy failed (logs.xlsx conflict)	Live log file was tracked by git	logs.xlsx removed from git; backup-pull-restore steps documented
Overlapping mind map	Long labels and repeated terms	Text wrapping, de-overlap layout, no repeated terms, course name as root
Evaluation visible to everyone	Sensitive model details	AI-models page only for the main admin, behind its own password
Lost questions	No follow-up	Follow-up table: Open / Answered + note
Images in slides not searchable	Search uses text only	Image notebook (CNN + image index) as a first step — not solved yet

CHAPTER 4

User Interface

This chapter explains the interface in general, then each of the three workflows in its own section: **4.2 Student**, **4.3 Doctor** and **4.4 Admin**. A permission matrix (4.5) compares them.

4.1 Common elements



- One sign-in page for all roles; the role is detected from the account.
- Guests receive a temporary code and can try the site immediately.
- Eye button on password fields and a strong-password generator on register.
- Language (EN/AR) and light/dark switch on every page.

Figure 17. Start page: sign in, register or continue as guest.

Table 6. Elements shared by all roles

Element	Description
Icon navigation capsule	Round icons with an animated tooltip that names the page and explains it; the active page is filled.
Account chip	Avatar, name and role; opens the profile (name, e-mail, photo, password, delete account).
Tools	Support 🛟, calendar 📅, notifications bell 🔔 with unread badge, dark mode, language.
Floating help	❓ on every page opens Help/FAQ (everyone reads; only staff add).
Course drop-down	Icon list of the user's courses with a check on the current one and search when there are many.
Date and live clock	Shown in the visitor's own time, ticking every second.



Figure 18. Arabic interface (right-to-left).



Figure 19. Dark mode.

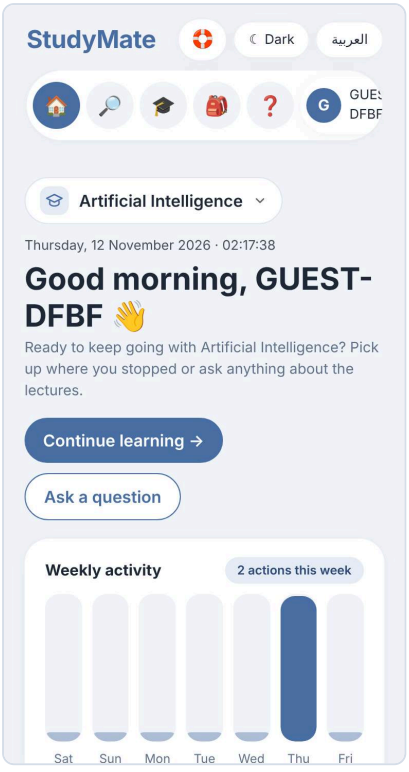


Figure 20. Phone: home

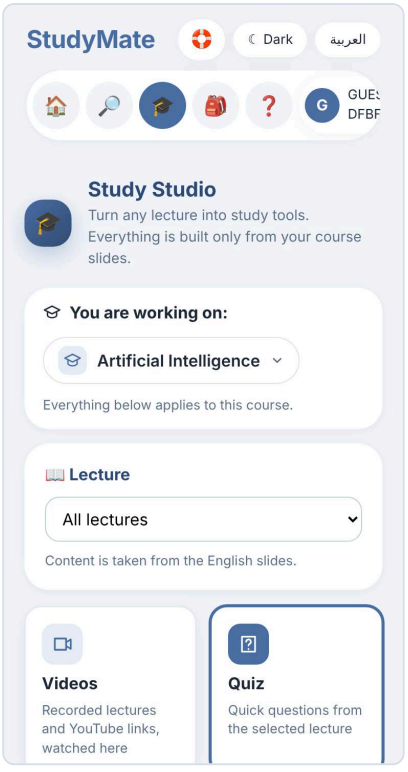


Figure 21. Phone: Studio

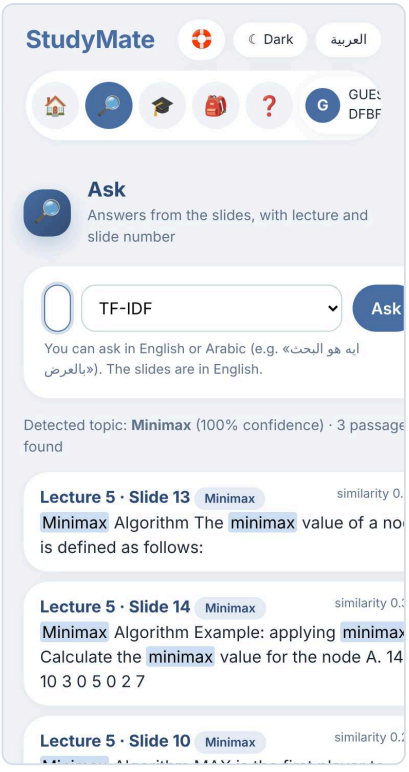


Figure 22. Phone: Ask

4.2 Student workflow

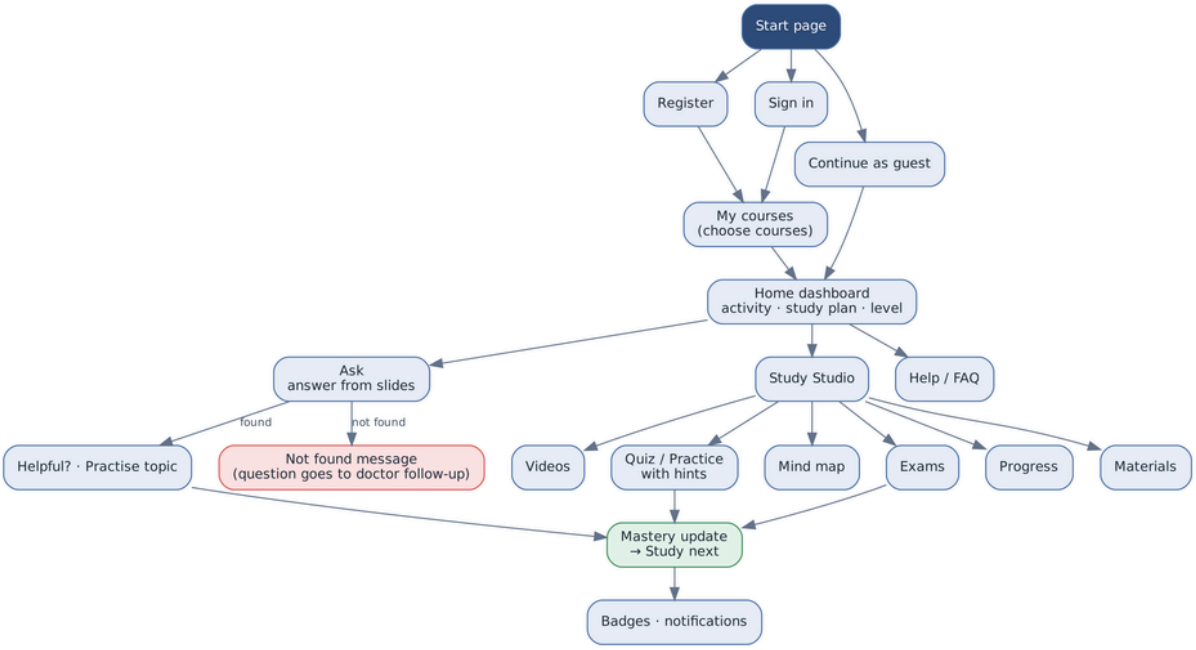


Figure 23. Student workflow: from the start page to answers, study tools and the mastery loop.

Table 7. Student workflow step by step

Step	What the student does	What the system does
1	Registers (or continues as guest) and chooses his courses	Creates the account (hashed password); notifies the admin and the course doctor
2	Opens Home	Shows weekly activity, overall level, study plan and "continue where you stopped"
3	Asks a question in English or Arabic and picks a search method	Returns the top-3 slides with lecture, slide, highlight, topic and confidence — or "Not found"
4	Marks the answer helpful / not helpful, or practises the topic	Logs it to logs.xlsx; the question appears in the doctor's follow-up
5	Uses Study Studio: videos, quiz, mind map, exams, progress, materials	Builds everything from the slides; updates the level per topic
6	Checks progress and badges	Colours topics red / amber / green and suggests what to study next

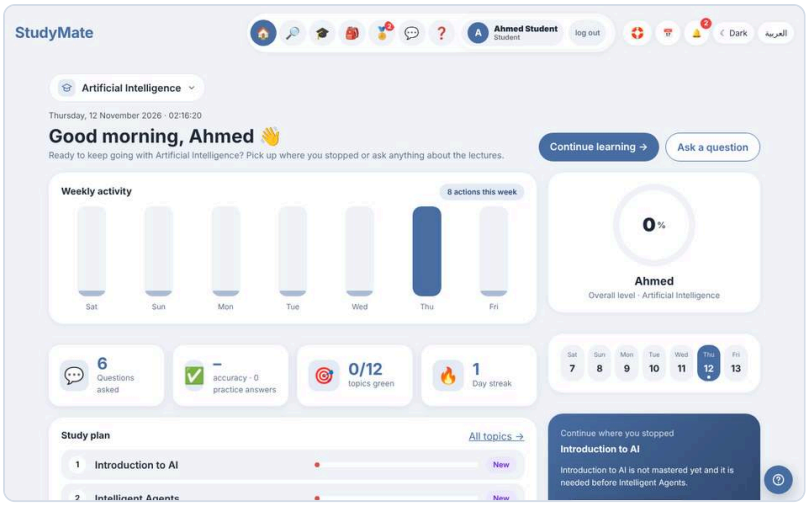


Figure 24. Student home

Student home

Greeting with date and live clock, weekly activity chart, overall level ring, stat cards (questions, accuracy, topics green, day streak), study plan and a "continue where you stopped" card.

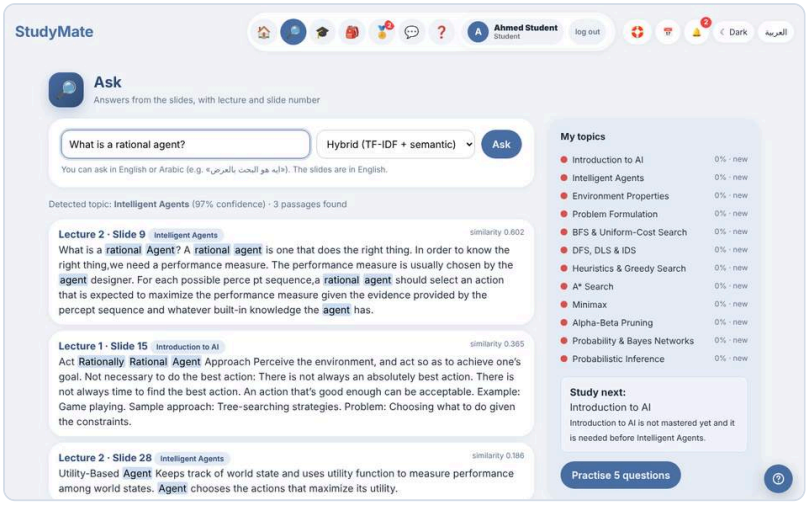


Figure 25. Ask: answer from the slides

Ask: answer from the slides

The three best slides with lecture and slide number, matching words highlighted, similarity score, detected topic and confidence. "Was this helpful?" and a button to practise the topic. The side panel shows the student's topics and what to study next.

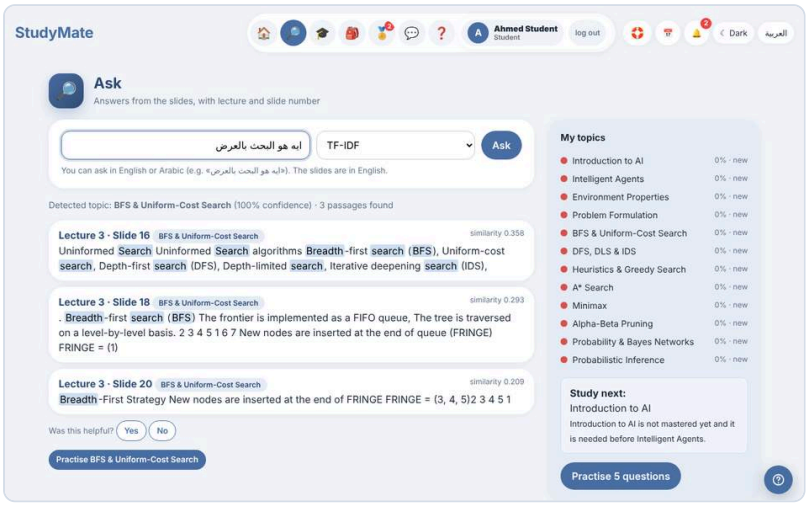


Figure 26. Arabic question

Arabic question

The Arabic question is translated with the course glossary and matched to the English slides.

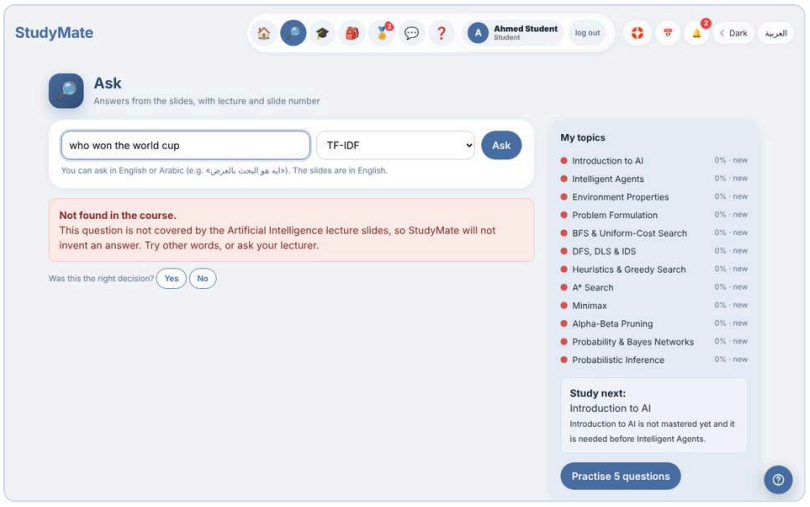


Figure 27. Not found in the course

Not found in the course

A question outside the slides is refused instead of inventing an answer; the doctor still sees it in the follow-up table.

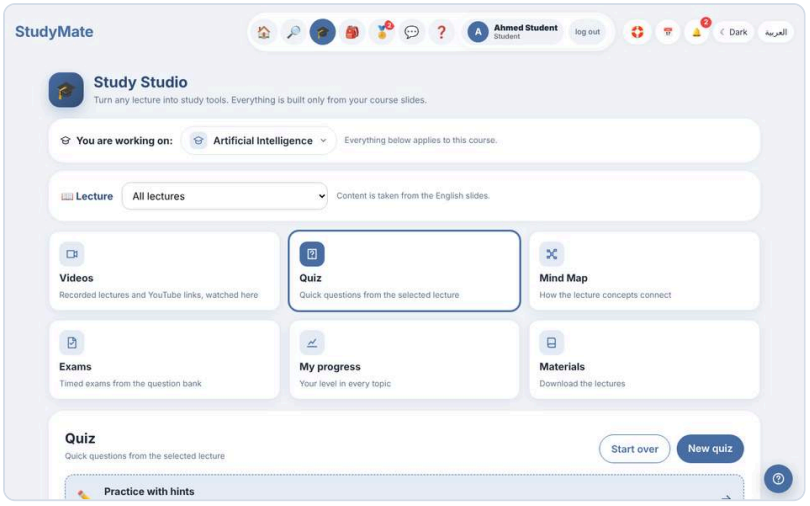


Figure 28. Study Studio

Study Studio

One place for all study tools of the selected course and lecture: Videos, Quiz, Mind map, Exams, My progress, Materials.

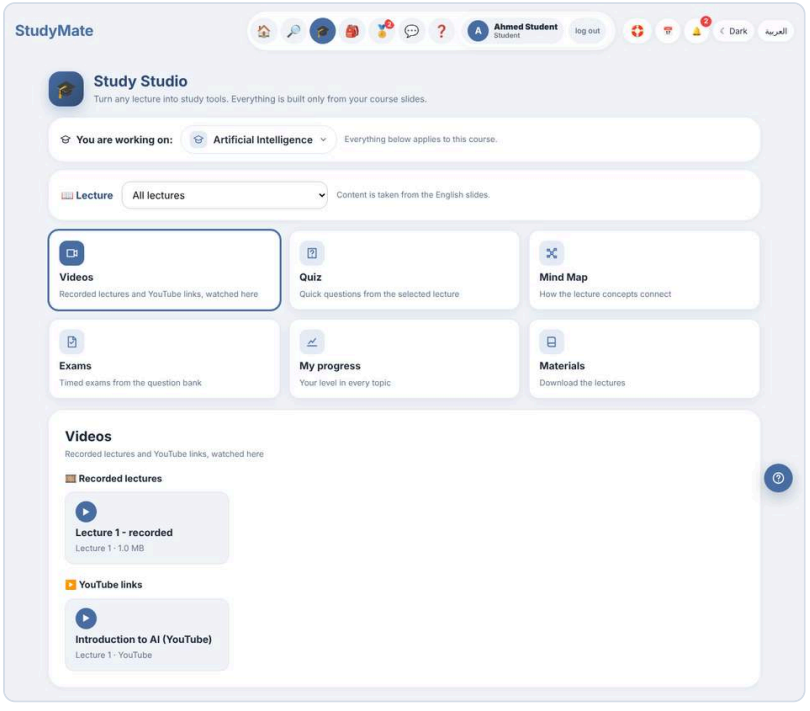


Figure 29. Videos

Videos

Two parts: recorded lectures uploaded by the doctor (HTML5 player) and YouTube links that play inside the page (no leaving the site).

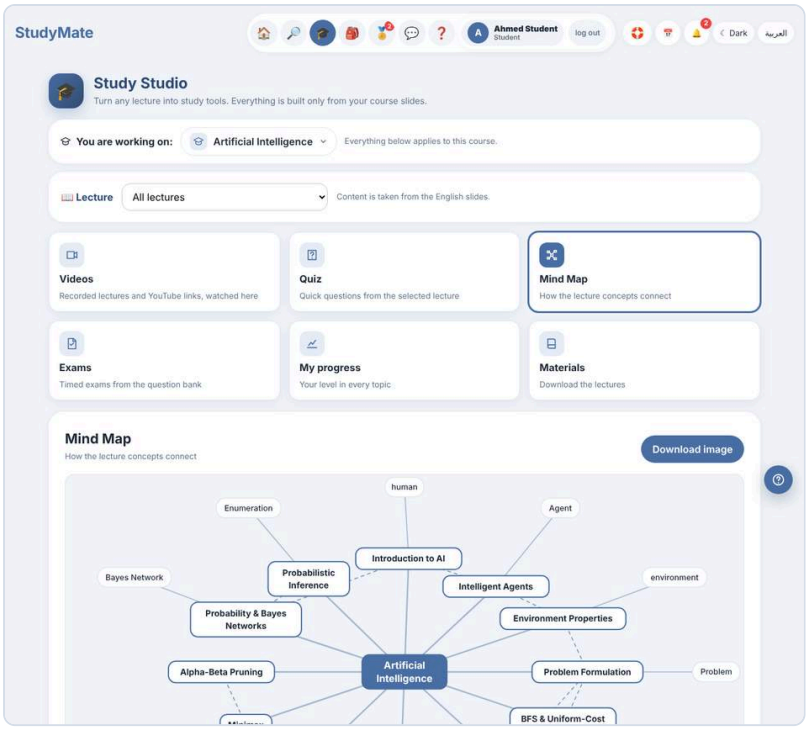


Figure 30. Mind map

Mind map

Key terms of the lecture grouped by topic around the course name; click a node to see the slides.

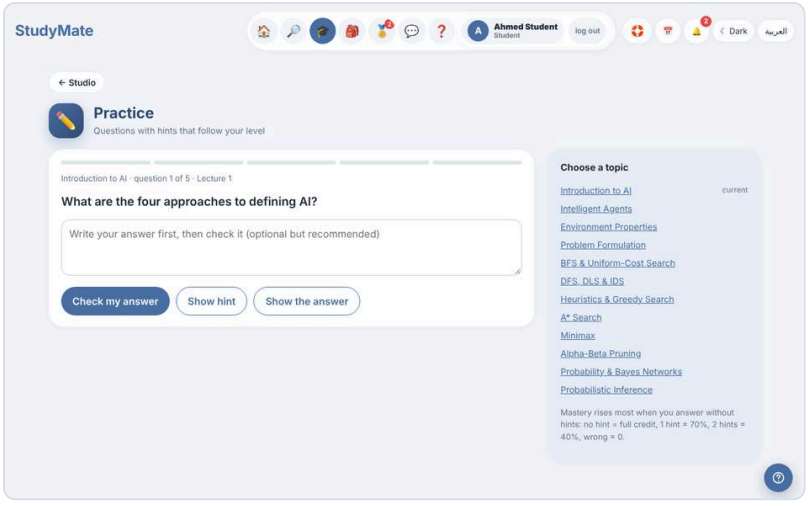


Figure 31. Practice with hints

Practice with hints

One question at a time; the student can ask for hint 1, hint 2 and the answer. Fewer hints = higher mastery gain.

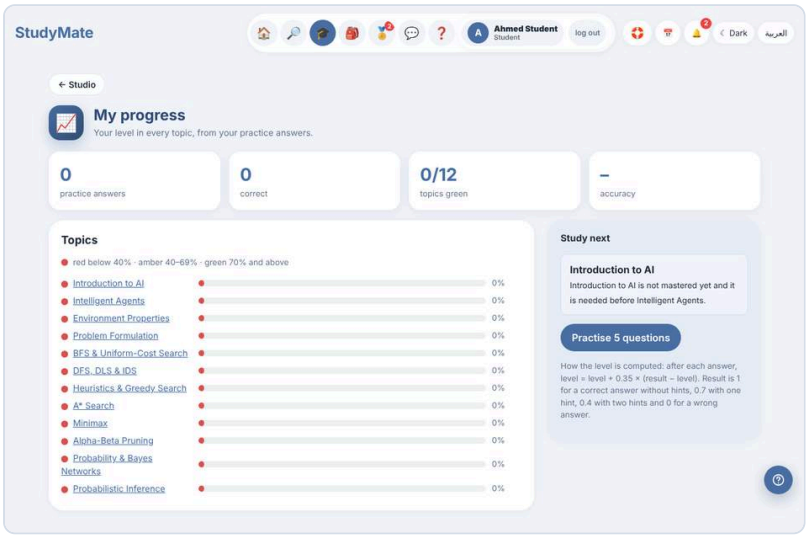


Figure 32. My progress

My progress

Every topic with its level (red < 40 %, amber 40–69 %, green ≥ 70 %) and a "study next" recommendation that respects the order of the topics.

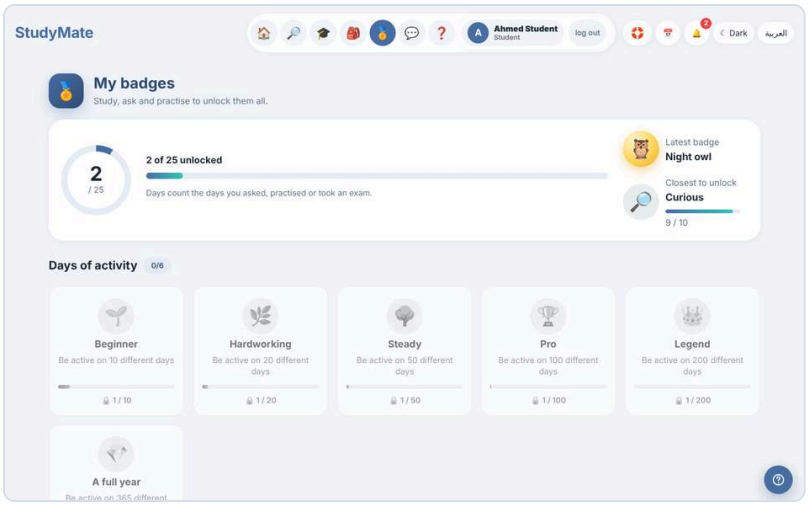


Figure 33. Badges

Badges

25 badges for active days, streaks, questions, practice, topics, exams and habits, plus custom badges from the doctor.

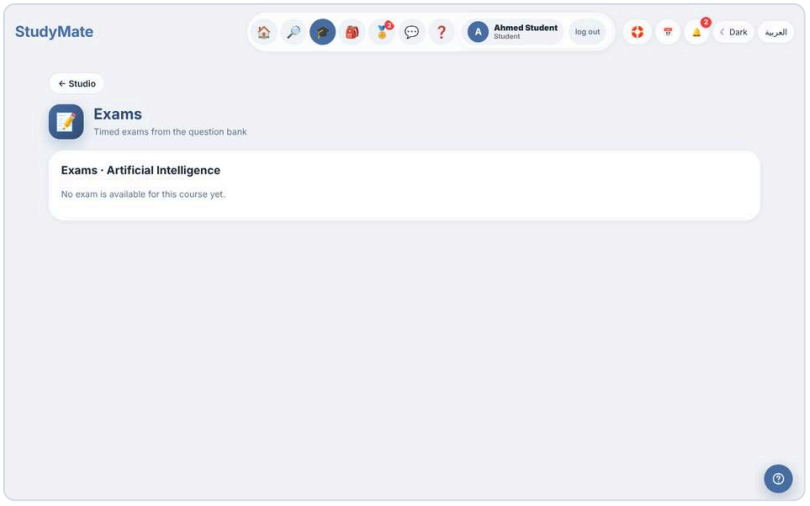


Figure 34. Exams

Exams

Timed exams from the doctor's question bank or a ready exam; MCQ / true-false graded automatically, essays by similarity; results stay private.

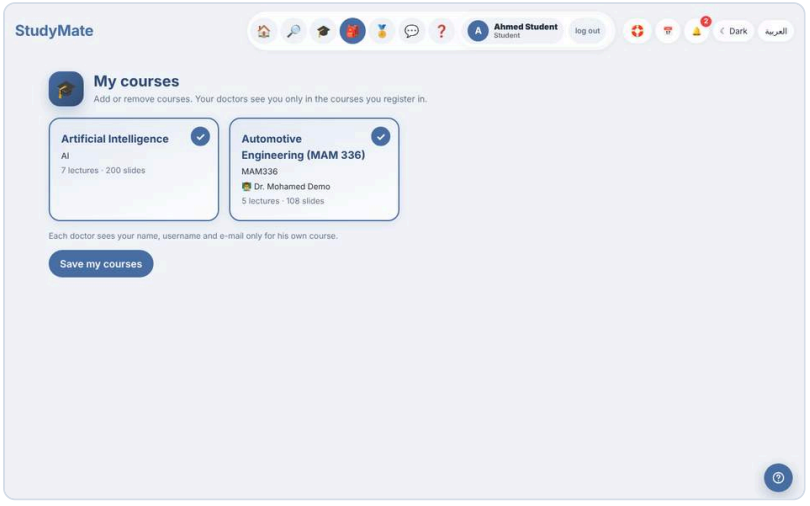
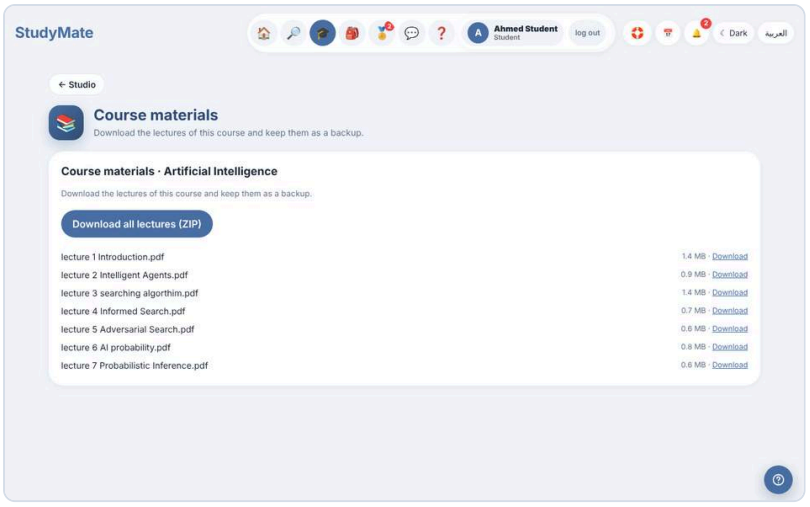


Figure 35. My courses

My courses

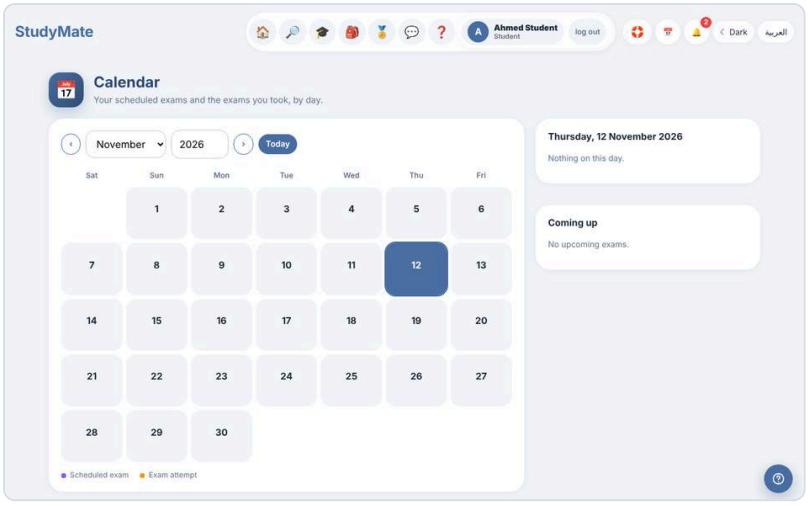
The student chooses or removes courses; each doctor then sees him among his students.



Materials

Download each lecture PDF or all of them as one ZIP.

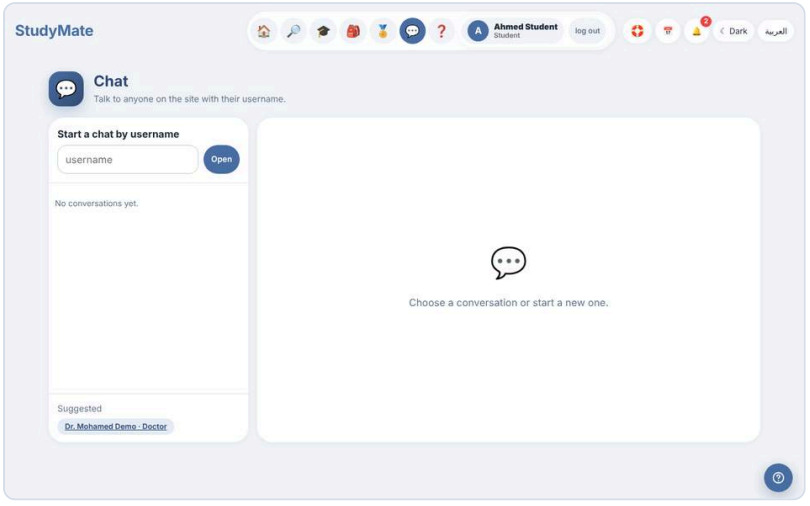
Figure 36. Materials



Calendar

Scheduled exams and the student's attempts by month and day.

Figure 37. Calendar



Chat

Direct messages between students and doctors by username.

Figure 38. Chat

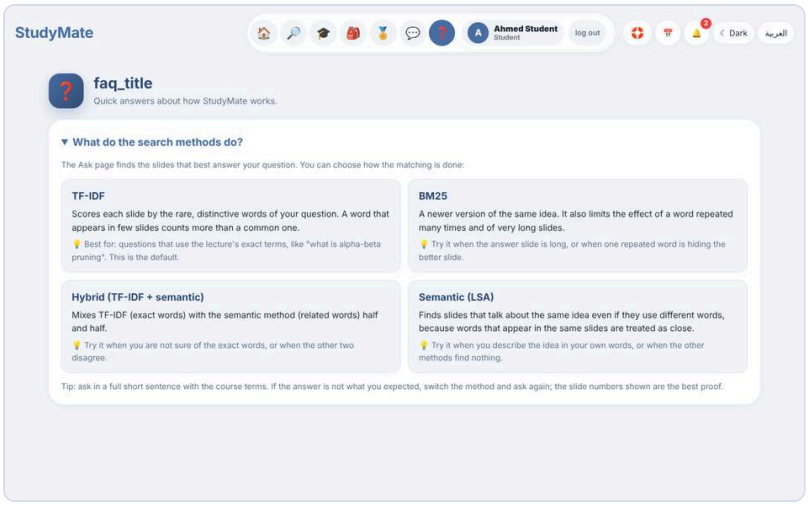


Figure 39. Help / FAQ

Help / FAQ

Explains the search methods and common questions; reachable from every page.

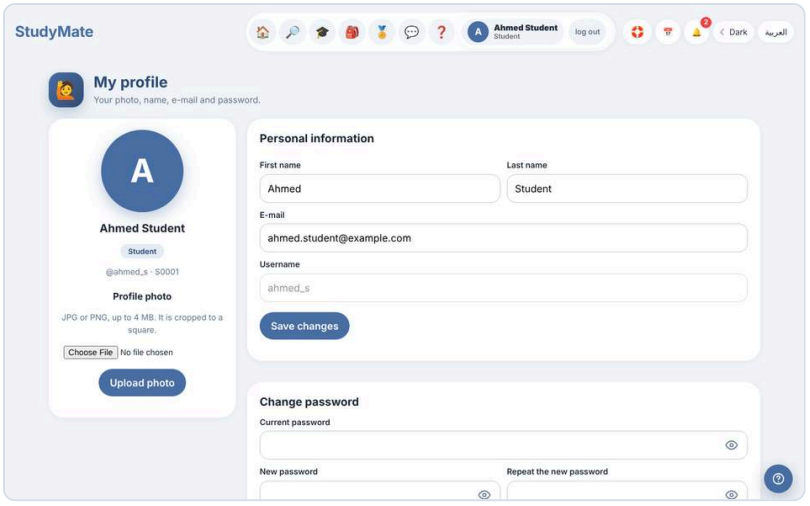


Figure 40. Profile

Profile

Name, e-mail, photo, password and "delete my account" (asks for the password, removes the student's data).

4.3 Doctor workflow

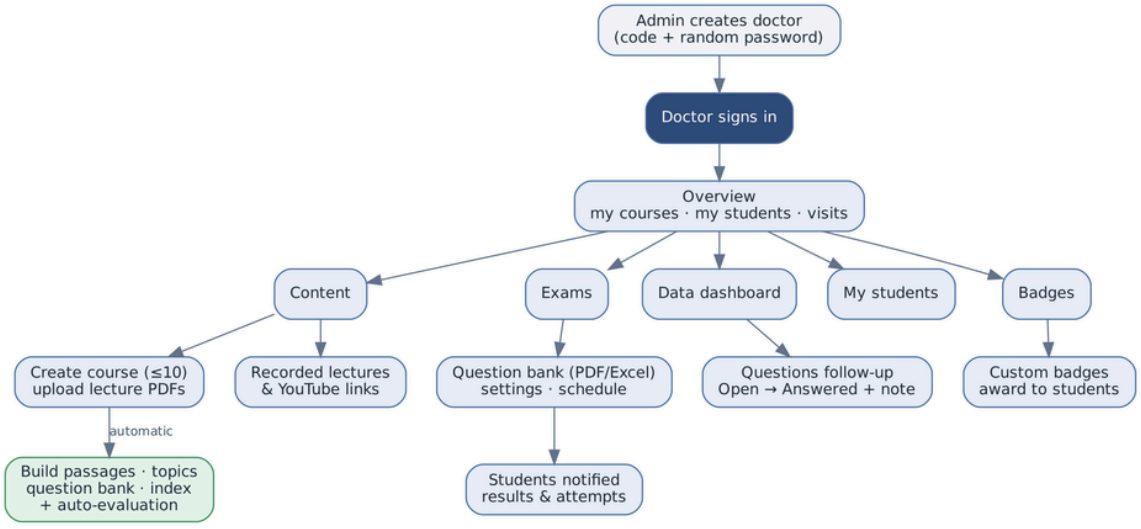


Figure 41. Doctor workflow: the admin creates the account; the doctor manages only his own courses.

A doctor account is created by the admin (code + random password, changeable later). A doctor works **only on the courses registered under his name** (up to 10) and sees **only the students registered in those courses**. He never sees other doctors' names, the site-wide numbers, the support inbox or the AI-models page.

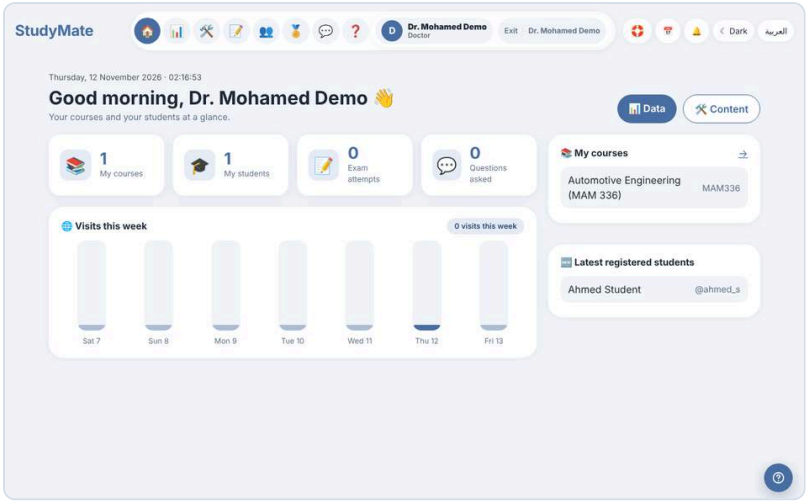


Figure 42. Doctor home

Doctor home

His courses, his students, exam attempts and questions asked; visits this week; latest registered students.

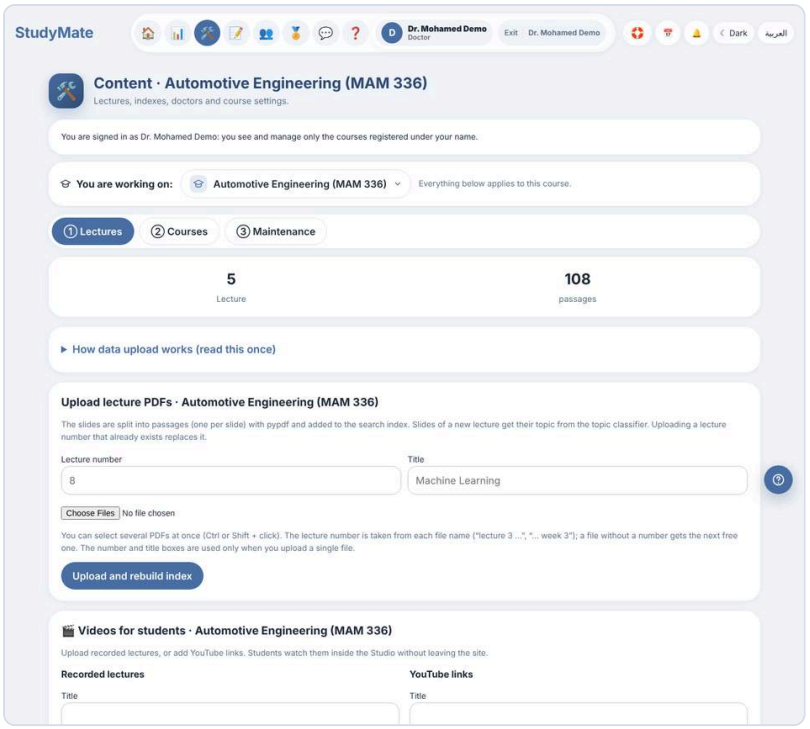


Figure 43. Content: lectures, courses, maintenance

Content: lectures, courses, maintenance

Upload one or many lecture PDFs (a new course is built automatically: passages, topics, glossary, question bank, search index, evaluation), delete lectures, rebuild the index or retrain; add recorded lectures and YouTube links for the Studio.

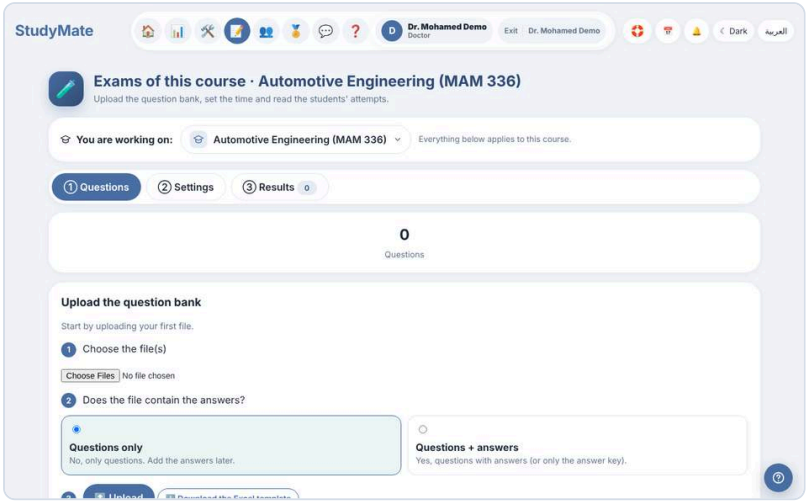
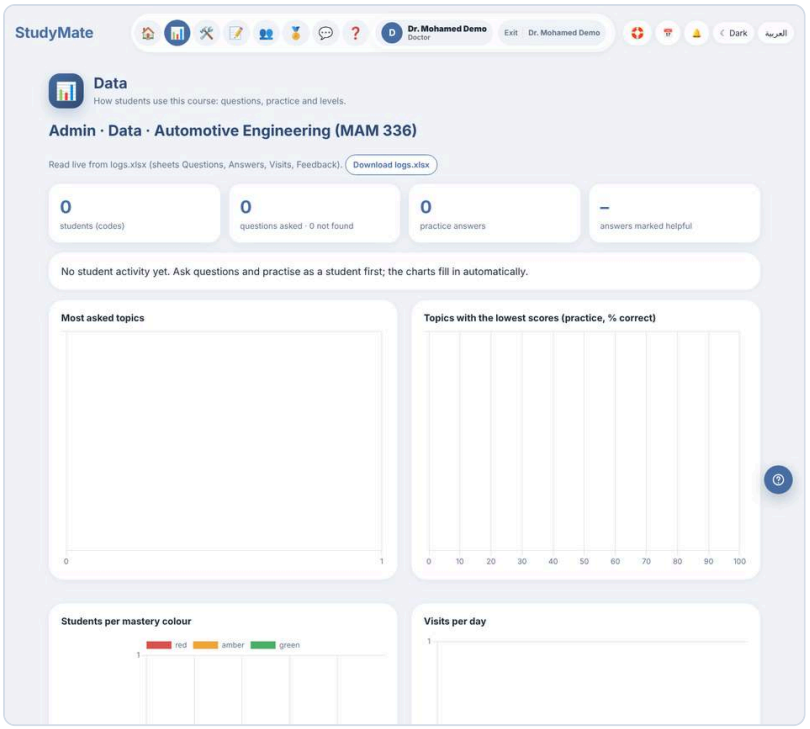


Figure 44. Exam manager

Exam manager

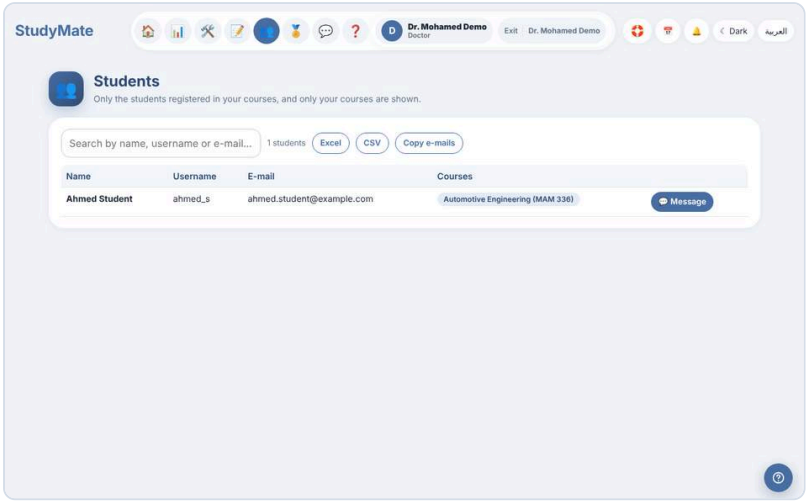
Questions / Settings / Results tabs: upload a question bank (PDF or Excel template), choose timings, schedule an exam (students are notified), review and export attempts.



Data dashboard and questions follow-up

KPIs and charts of his course; the follow-up table lists the latest questions with Found / Not found and a status — the doctor marks a question **Answered** with a note or reopens it.

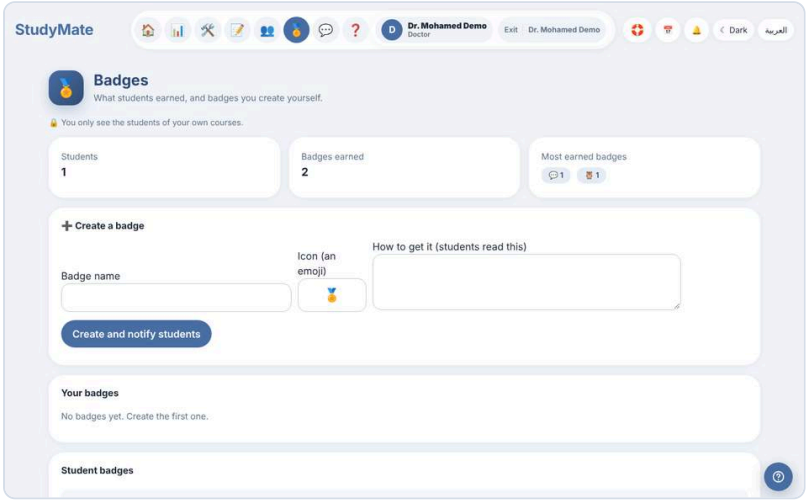
Figure 45. Data dashboard and questions follow-up



My students

Only students registered in his courses, with search and Excel/CSV export; a Message button opens a chat.

Figure 46. My students



Badges management

Sees the badges his students earned and creates custom badges to award them (students get a notification).

Figure 47. Badges management

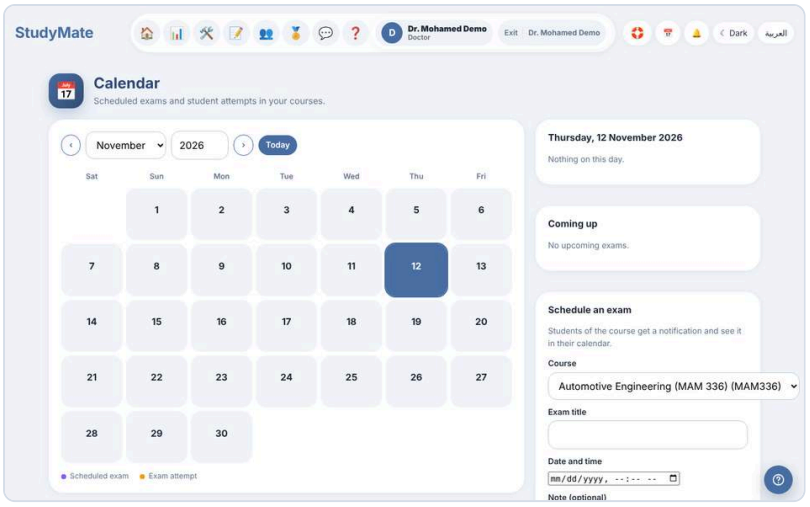


Figure 48. Calendar

Calendar

Schedules exams for his courses; sees attempts per day.

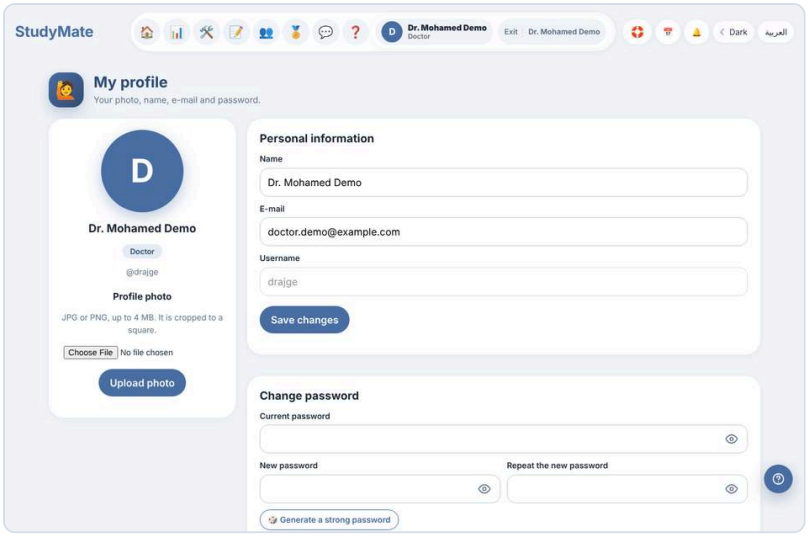


Figure 49. Profile

Profile

Changes his own name, e-mail, photo and password.

4.4 Admin workflow

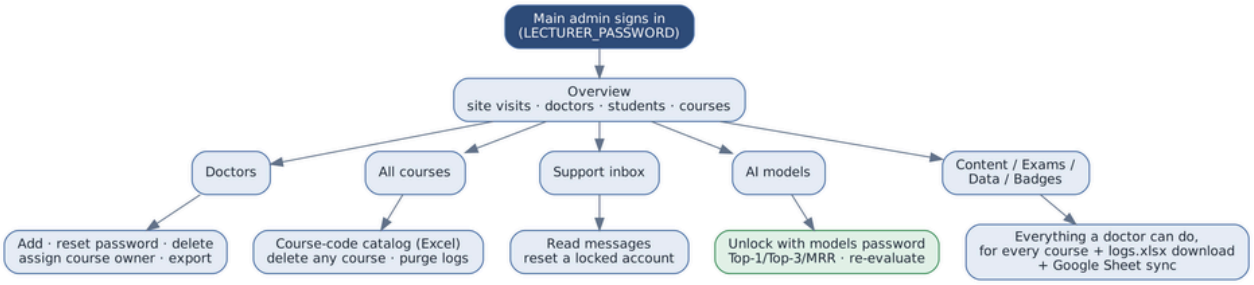
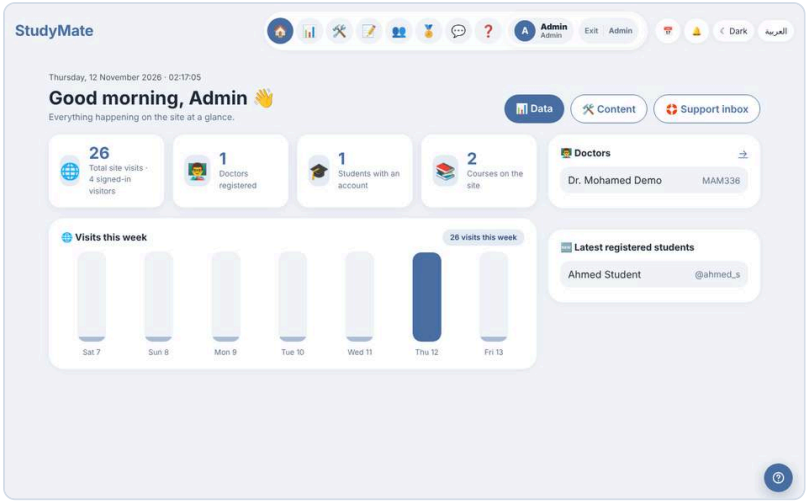


Figure 50. Admin workflow: the main admin controls doctors, all courses, support and the AI models.

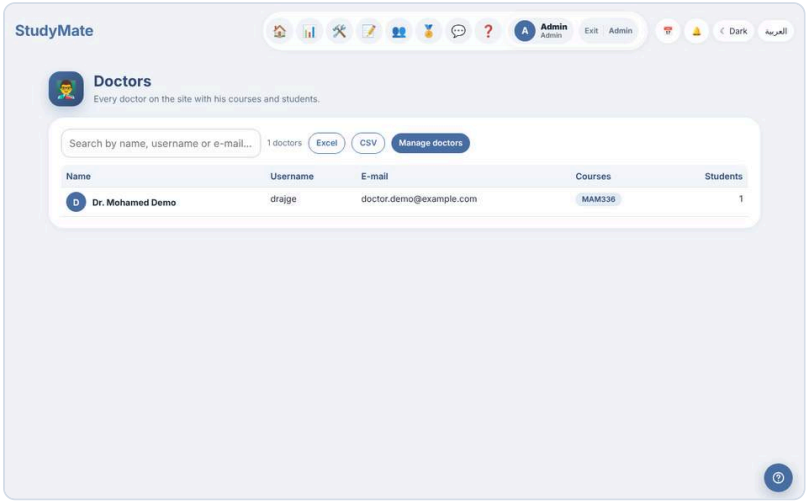
The main admin signs in with the password set on the server (`LECTURER_PASSWORD`). He can do everything a doctor can do, for **every** course, plus the site-wide management below.



Admin home

Total site visits, doctors, students with an account, courses; weekly visits; the doctors and their courses; latest students; shortcuts to Data, Content and the Support inbox.

Figure 51. Admin home



Doctors

List with search and Excel/CSV export. From Content ▸ Courses the admin adds a doctor (random password shown once), resets his password, deletes him or assigns a course owner.

Figure 52. Doctors

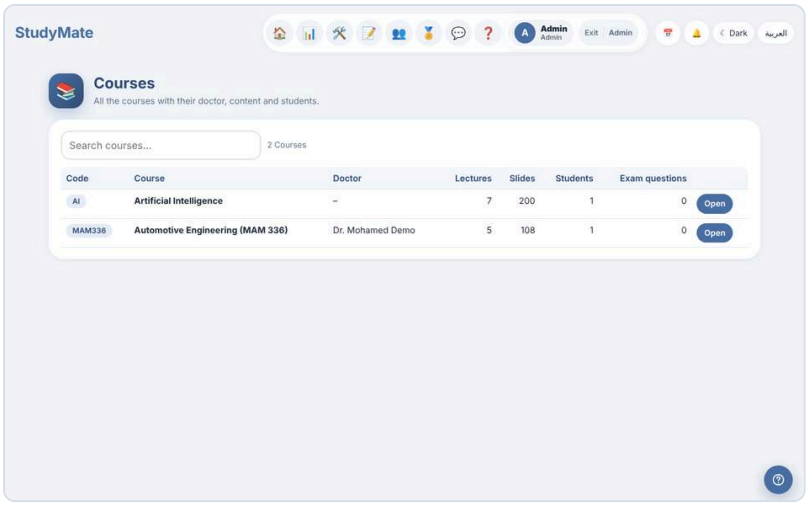


Figure 53. All courses

All courses

Every course with its doctor, lectures, slides and students; the course-code catalog (Excel) feeds the course drop-down; any course can be deleted (optionally with its logs).

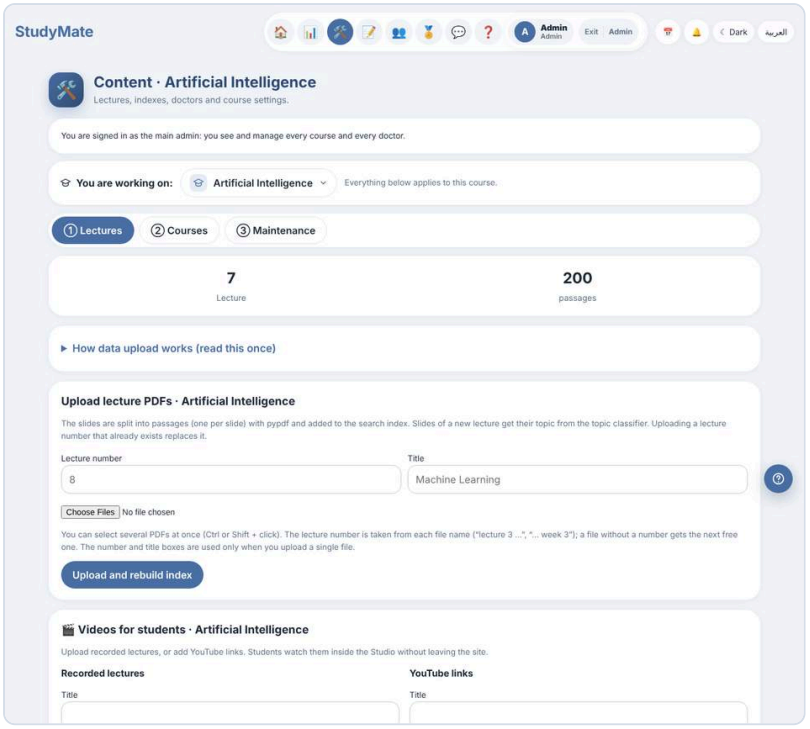


Figure 54. Content (all courses)

Content (all courses)

Same tools as the doctor for any course, plus doctor management, the course-codes list and the optional Google Sheet sync.

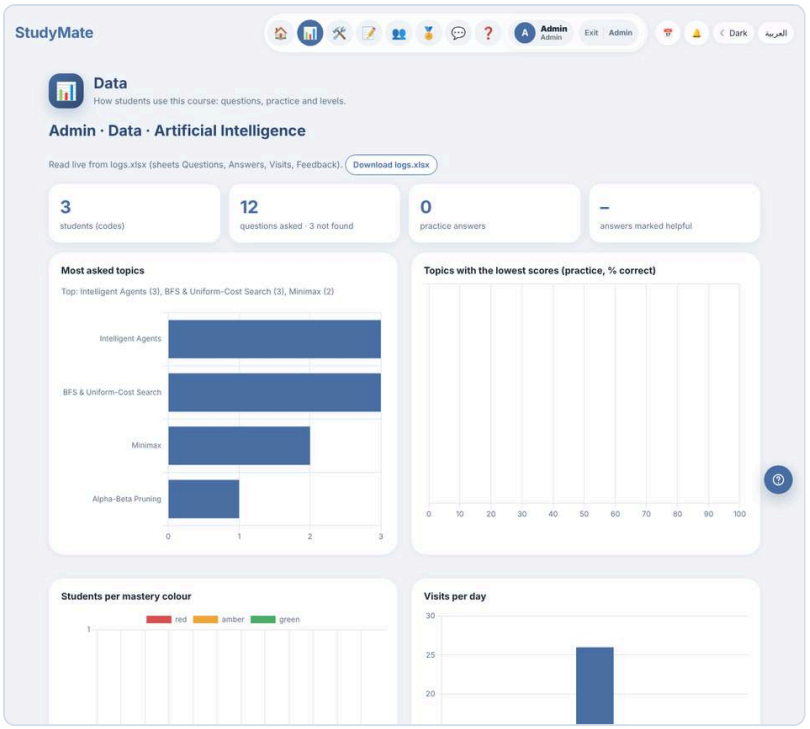


Figure 55. Data dashboard (all students)

Data dashboard (all students)

KPIs, most asked topics, weakest topics and the full questions follow-up; download the complete logs.xlsx.

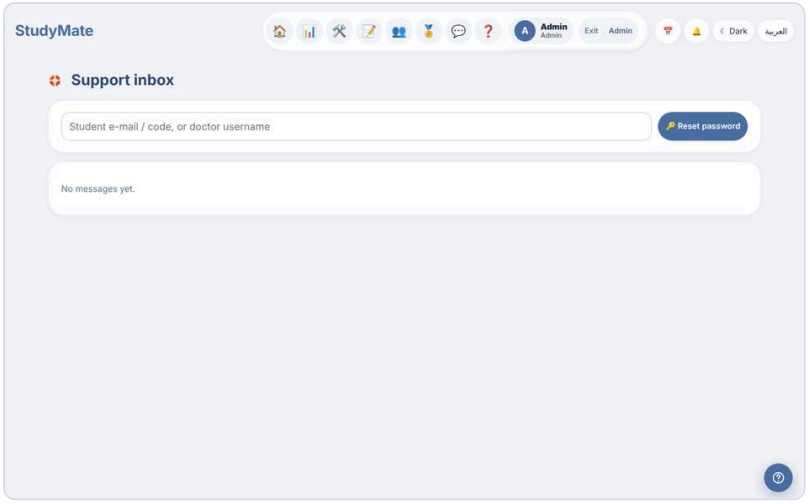


Figure 56. Support inbox

Support inbox

Messages sent with the 🛟 button (also from locked-out users); the admin can reset a student's or doctor's password — the new password is shown once and unlocks the account.

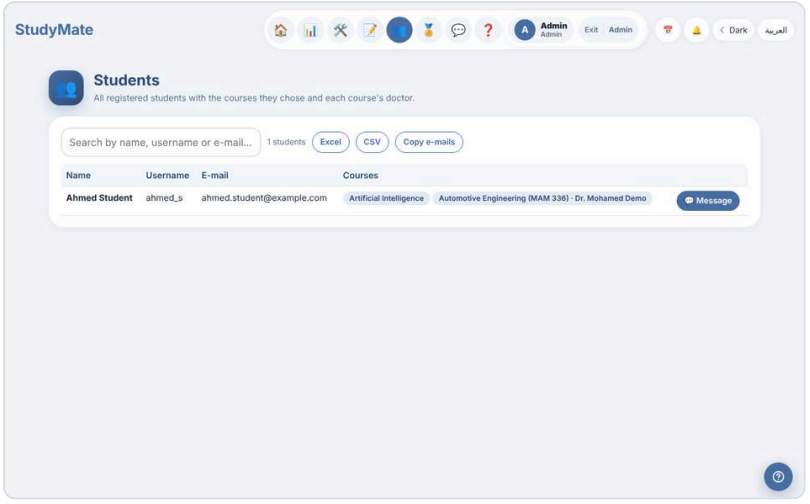


Figure 57. All students

All students

Every student with courses and activity; Excel/CSV export.

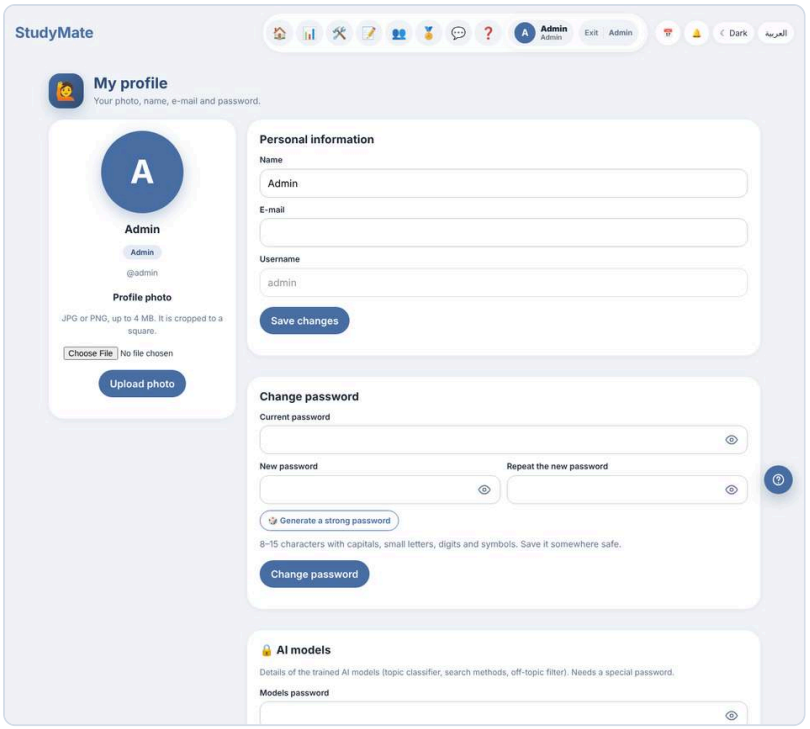


Figure 58. Profile and AI-models unlock

Profile and AI-models unlock

The main admin's profile contains the AI-models card: unlock with its own password, lock again, change that password.

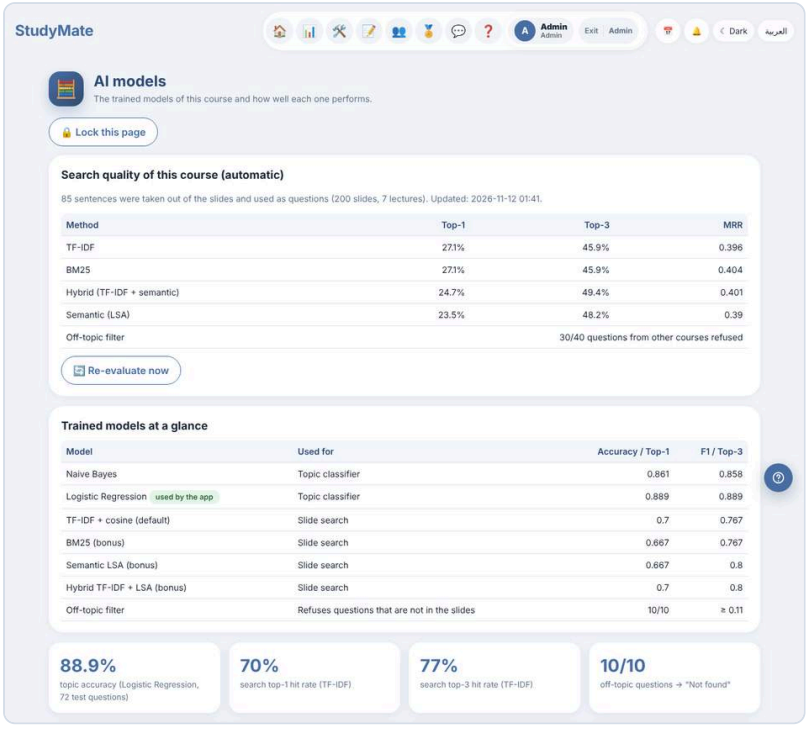


Figure 59. AI models page: automatic evaluation table and history.

AI models (main admin only)

Top-1 / Top-3 / MRR of every search method for the selected course, off-topic test, history of past runs and a **Re-evaluate now** button. Below it: the trained classifiers, confusion matrices, search test, robustness and Arabic examples.

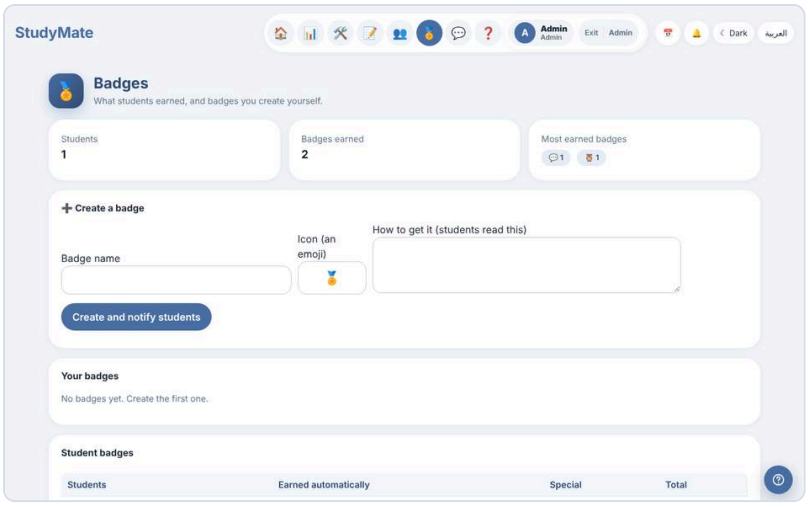


Figure 60. Badges of all students.

Badges

Same as the doctor's page but for every student and every custom badge.

4.5 Permission matrix

Table 8. Who can do what (enforced in app.py by student_required, lecturer_required, is_master and can_manage)

Action	Student	Doctor	Admin
Ask questions, Studio, practice, exams, progress, badges	✔	—	—
Read Help/FAQ, chat, calendar, notifications, profile	✔	✔	✔
Add / delete FAQ questions	—	✔	✔
Choose own courses / delete own account	✔	—	—
Create a course, upload / delete lectures, rebuild, retrain	—	own courses	✔
Upload recorded lectures / YouTube links	—	own courses	✔
Exam bank, settings, schedule, results	—	own courses	✔
Data dashboard and questions follow-up	—	own courses	✔
See students / export	—	own courses	✔
Custom badges	—	own courses	✔
Site overview (all visits, doctors, students)	—	—	✔
Add / reset / delete doctors, assign course owner	—	—	✔
All courses, course-code catalog, delete any course	—	—	✔
Download full logs.xlsx, Google Sheet sync	—	—	✔
Support inbox and password reset	—	—	✔
AI-models page (password-protected)	—	—	✔

> Areas are separated: when a doctor or the admin is signed in, student pages redirect to the staff home, and student accounts can never open staff pages. A doctor's rights are checked on every action with `can_manage(course)`.

CHAPTER 5

Model Training Notebooks (Google Colab)

Table 9. Colab notebooks (links in the README)

Notebook	Content	Output
StudyMate_Search_Models.ipynb	Clones the repo, uses the site's own code; trains and evaluates TF-IDF / BM25 / Hybrid / LSA and the topic classifier on any course; optional upload of your own PDFs	accuracy tables, charts, auto_metrics.json, ZIP for the TA
StudyMate_Image_Recognition.ipynb	Extracts the pictures from the lecture PDFs, removes logos/duplicates, trains a CNN (MobileNetV2 on Colab), builds an image→slide search index and tests it with distorted "phone photos"	image_model.keras, image_index.npz, metrics, figures

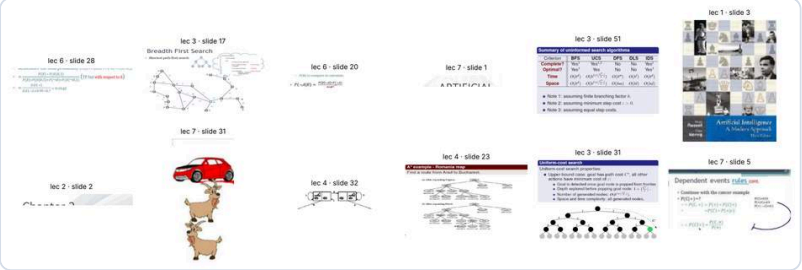


Figure 61. Pictures extracted automatically from the AI lecture PDFs.

292 pictures were extracted from the 7 AI lectures; after removing logos and duplicates 250 remain. Each keeps its lecture and slide number so that a match can point the student to the slide.

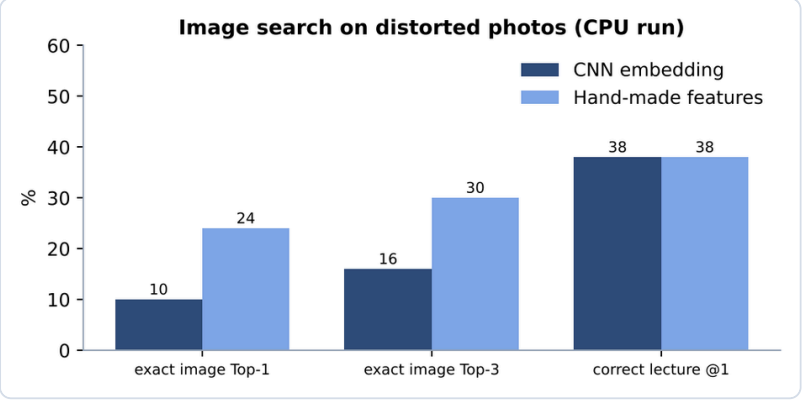


Figure 62. Image search on distorted copies of test images (offline CPU run).

In the offline run ImageNet weights could not be downloaded, so a small CNN was trained from scratch: lecture accuracy 26 % and a simple hand-made feature beat it for image search. With so few, similar pictures this is expected. On Colab the notebook uses MobileNetV2 transfer learning — **that run still has to be done**; image search is therefore not yet part of the website.

CHAPTER 6

Testing, Security and Deployment

Table 10. Quality and security

Area	What was done
Automated tests	70 pytest tests (search, not-found, Arabic, practice/mastery, exams, roles, studio, videos, follow-up, auto-evaluation, help, neural fallback) — all passing on a clean copy
Passwords	Hashed (scrypt); 5 attempts then lock; strong generator; models page has its own hashed password
Privacy	Students identified by a code; accounts, logs, videos and passwords are git-ignored and never committed; a student can delete his account and data
Secrets	LECTURER_PASSWORD, SECRET_KEY, STUDYMATE_MODELS_PASSWORD and mail keys come from environment variables (set in the WSGI file)
Uploads	Type checks (PDF / MP4, WebM, OGG), size limit, files served only to signed-in students
Hosting	PythonAnywhere (free): git pull + Reload; renew monthly; guides for VPS, Render and Docker in docs/DEPLOY.md

CHAPTER 7

Conclusion, Limitations and Future Work

7.1 Conclusion

StudyMate meets every requirement of Project 1E and extends it into a usable, deployed platform. The central design decision — answering with real slides instead of generated text — keeps every answer verifiable, and the rejection gate prevents answers outside the course. The evaluation shows that simple, well-tuned classical models (TF-IDF, LSA, Logistic Regression) are accurate on this task, run on free hosting, and can be re-evaluated automatically for every new course. The three-role design lets students study, lecturers follow their students and the administrator manage the whole site with clearly separated permissions.

7.2 Limitations

- Pictures inside slides are not searchable yet (image notebook is the first step).
- Scanned PDFs depend on OCR quality (MAM336); its question bank has no answers yet.
- Very short questions (two words) are hard to match.
- Free hosting: outbound allow-list (no neural model download, no mail API), 512 MB disk, monthly renewal.
- logs.xlsx suits a class, not thousands of users.

7.3 Future work

- Run the image notebook on Colab and add "search by picture" to Ask.
- Enable the neural method on a host with open internet; compare it in the evaluation.
- Move logs to SQLite/PostgreSQL; e-mail notifications.
- A formal user study with the course lecturer and a group of students, as planned in the project sheet.

APPENDIX A

Deliverables and How to Run

Table 11. What is handed in

Item	Where
Source code (latest version)	GitHub ▸ Code ▸ Download ZIP of the main branch
Trained models, data, data card, test sets	`models/`, `data/AI/`, `static/reports/`
Colab notebooks	`notebooks/`
This report and the presentation outline	`docs/`
Live system	https://ahmedhallawa.pythonanywhere.com
Never included	accounts (students/doctors/admin JSON), logs.xlsx, secrets/, real passwords

Running the system locally

```
git clone https://github.com/pikachu8240/StudyMate.git && cd StudyMate
python -m venv venv && source venv/bin/activate      # Windows: venv\Scripts\activate
pip install -r requirements.txt
export LECTURER_PASSWORD=...  SECRET_KEY=...
python app.py                                        # http://127.0.0.1:5000
python -m pytest -q                                  # 70 tests
```

Repository structure

Table 12. Repository layout

Path	Content
`app.py`	Flask application: 64 routes, roles and sessions
`config.py`	Settings, course paths, time zone, environment variables
`core/`	Search, classifier, mastery, studio, exams, accounts, videos, evaluation, i18n
`scripts/`	build_passages.py, train.py, evaluate.py
`data/<course>/`	Lecture PDFs, passages.csv, topics, question bank, labelled questions, test sets, glossary, data card
`models/`	Trained search index and classifiers (joblib)
`static/reports/`	Metrics, confusion matrices, threshold plot
`templates/`, `static/`	Jinja pages, CSS themes, JavaScript
`notebooks/`	Google Colab notebooks and their results
`tests/`	70 automated tests (pytest)
`docs/`	README screenshots, deployment and hand-over guides, this report

APPENDIX B

Complete Change Log

Table 13. All 64 commits in order

commit	date	change (git message)
245805f	2026-11-09	StudyMate: course study assistant (Flask) + Render deploy config
f26eb8e	2026-11-09	Arabic/English interface, dark mode, multi-course support
ef59e66	2026-11-09	Reset logs.xlsx (empty, with the course column)
a9a4f61	2026-11-09	Add Automotive Engineering course, guest mode, lecturer-only evaluation, admin help
fd4a544	2026-11-09	Admin: delete lecture / delete course; search-method hint on the Ask page
d114274	2026-11-09	Admin: upload several lecture PDFs at once
05fc34a	2026-11-09	Merge Lecturer and Admin into one Admin area; hide student pages while in Admin
2ee0c8f	2026-11-09	Remove course name next to the site name in the header
8621542	2026-11-09	Doctor accounts with private codes, per-doctor courses and data, course search drop-down
a9d93c7	2026-11-09	Start page: course picker is a searchable drop-down
f591514	2026-11-09	Start page: type directly in the course field to search
cefff09	2026-11-09	README and presentation outline: final version
fa33d12	2026-11-09	Data card: glossary count
84fab52	2026-11-10	Doctors: random password, own password change, private Google Sheet sync (admin only)
6559ad7	2026-11-11	Admin: hide Google Sheet connection card until key file or SHEET_SYNC=1
bd3ba64	2026-11-11	Accounts (username/password, register, forgot, remember me), exams, Arabic translation, materials, admin exports
7767081	2026-11-11	Exams: ignore PDF headers/footers, accept an answer-key-only file
24f8191	2026-11-11	Fix the new exam parser test
b860591	2026-11-11	Student home dashboard, UI polish, exams: live language, essay similarity grading, results history
dcc1ce2	2026-11-11	Students choose their courses; each doctor sees only his own students; site-wide UI polish
b8d3276	2026-11-11	Edu theme: light dashboard UI, capsule icon nav, new student home (activity chart, profile ring, week calendar, study pl
5150f95	2026-11-11	Doctors sheet section: English only (table headers, title, notes)

commit	date	change (git message)
6fdf911	2026-11-11	Admin/doctor: delete several lectures (or all) of a course in one step
2cb1e79	2026-11-11	Admin/doctor home page: doctors, students, traffic, courses (doctor: courses, students, exams, questions); students card
4d220b0	2026-11-11	Remove page lag: async visit logging, cached Excel reads, faster fade-in, no hover-expanding nav, no web-font import
ce81790	2026-11-11	Add calendar with exam scheduling, notifications bell, profile page with photo, doctors and courses pages; reorder admin
97919ee	2026-11-11	Remove the course drop-down from the top bar; students switch course with chips on their home page
63b9876	2026-11-11	Semantic search (LSA) and hybrid TF-IDF+LSA, selectable on Ask page and compared in Evaluation
ad9ff67	2026-11-11	Login: 5 tries per account with support pop-up (EN/AR); support icon + message form; admin support inbox
86bf941	2026-11-11	Strong password generator (trigram language model + secrets) on register and profile pages
b8464ba	2026-11-11	Forgot password: strong random password e-mailed with the login-details template (Brevo/Resend/SMTP), saved only after s
40feab6	2026-11-11	Nav icons: click shows an animated tooltip (name + what it does), AR/EN
61a79b1	2026-11-11	Admin password reset from the support inbox (shown once, works at once, unlocks); icon tooltips on hover
094e834	2026-11-11	Eye button on every password field; generated password fills the fields (shown) instead of a separate line
3b61c2d	2026-11-11	Remove the footer line everywhere and the main-admin password note from the profile page
e280bfd	2026-11-11	Badges (25, days/streak/questions/practice/topics/exams/habits) with page, nav icon and bell notices; mistakes review mo
5665852	2026-11-11	Delete any course (even AI / all of them); course-code dropdown fed by an Excel list the admin uploads; no-courses page
75137d6	2026-11-11	Admin/doctor badges page, custom badges with notifications, purge-logs on course delete
9b6e7fa	2026-11-11	Main admin can change own password from the profile; no support button for the admin
0cc8a97	2026-11-11	Course codes list: uploaded courses fill in names, new courses join the list, import button
5841e14	2026-11-11	Evaluation page is for the main admin only (hidden from doctors)
eba4ecf	2026-11-11	Exams page restructured into Questions / Settings / Results tabs (admin and doctors)
52766f2	2026-11-11	Home shortcut tiles (admin/doctor and student) in a single row on every screen size
c95cd77	2026-11-11	Exams page adapts to the uploaded questions: no essay chip, essay settings only when essay questions exist
bbc3592	2026-11-11	Content page restructured into Lectures / Courses / Maintenance / Account tabs (admin and doctors)
632f7cd	2026-11-11	Exam-finished notifications no longer open the student's attempt (privacy)
bb6680d	2026-11-11	Exam settings reduced to two times: multiple choice and true/false (2-option questions); essay settings removed from the
6c398db	2026-11-11	Course picker on the Content and Exams pages (admin and doctors)

commit	date	change (git message)
9ab6b74	2026-11-11	Header exit button reads "Exit" followed by the signed-in name
fae888f	2026-11-11	Remove "My account" tab from the Content page (password lives in the profile)
4628a6e	2026-11-11	Chat: direct messages on the site (by username), doctor/student Message button opens a chat, unread badge and notificati
c1f0c17	2026-11-11	Exams: "delete the question bank" moved from the Questions tab to Settings
b71c7a6	2026-11-11	Opening the site link from outside (typed address, bookmark, shared link) always asks for a sign-in
1fa66d4	2026-11-12	Study Studio: one student tab with a summary report, flashcards, quick quiz, terms table (+Excel), concept map, infograp
649494e	2026-11-12	Ready exam option, search-method explanations, Cairo date, remove all courses, delete account with password
37b48ad	2026-11-12	Studio: tidy mind map, stacked listen notes with EN/AR voice, practice merged into quiz, Exams/Progress/Materials moved
13d4119	2026-11-12	Studio shelf (quiz, mind map, slide deck + exams/progress/materials), slide preview, Help page with editable FAQ, My cou
71d9689	2026-11-12	Slate-blue palette, auto-evaluation of search models, questions follow-up, videos in Studio, live clock; stop tracking l
d1a85f1	2026-11-12	Add Colab notebooks: search-model training/evaluation and lecture-image recognition (with executed outputs)
905c01c	2026-11-12	Studio: remove slide deck, unify tile icons; course dropdown with icons (title only); floating Help on every page (publi
20738b3	2026-11-12	Optional free neural search method (fastembed, graceful fallback to Hybrid)
dc77931	2026-11-12	README with screenshots, DEPLOY and HANDOVER guides
2dba9a8	2026-11-12	Add Arabic project report (PDF)
8aaeccf	2026-11-12	Professional project report (English for submission, Arabic copy) with screenshots, workflows and permissions

APPENDIX C

References and Tools

1. C. D. Manning, P. Raghavan, H. Schütze, *Introduction to Information Retrieval*, Cambridge University Press, 2008 (TF-IDF, cosine similarity, evaluation).
2. S. Robertson, H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond", *Foundations and Trends in Information Retrieval*, 2009.
3. S. Deerwester et al., "Indexing by Latent Semantic Analysis", *Journal of the American Society for Information Science*, 1990.
4. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python", *JMLR* 12, 2011 (TfidfVectorizer, TruncatedSVD, LogisticRegression, MultinomialNB).
5. M. Sandler et al., "MobileNetV2: Inverted Residuals and Linear Bottlenecks", *CVPR*, 2018 (image notebook).
6. AI Lab Projects sheet, Project 1E — Course Study Assistant (project requirements).
7. Software: Python 3, Flask, Jinja2, pypdf, pandas, openpyxl, rank_bm25, wordninja, joblib, matplotlib, python-pptx, TensorFlow/Keras (Colab), fastembed (optional); hosted on PythonAnywhere; code on GitHub.